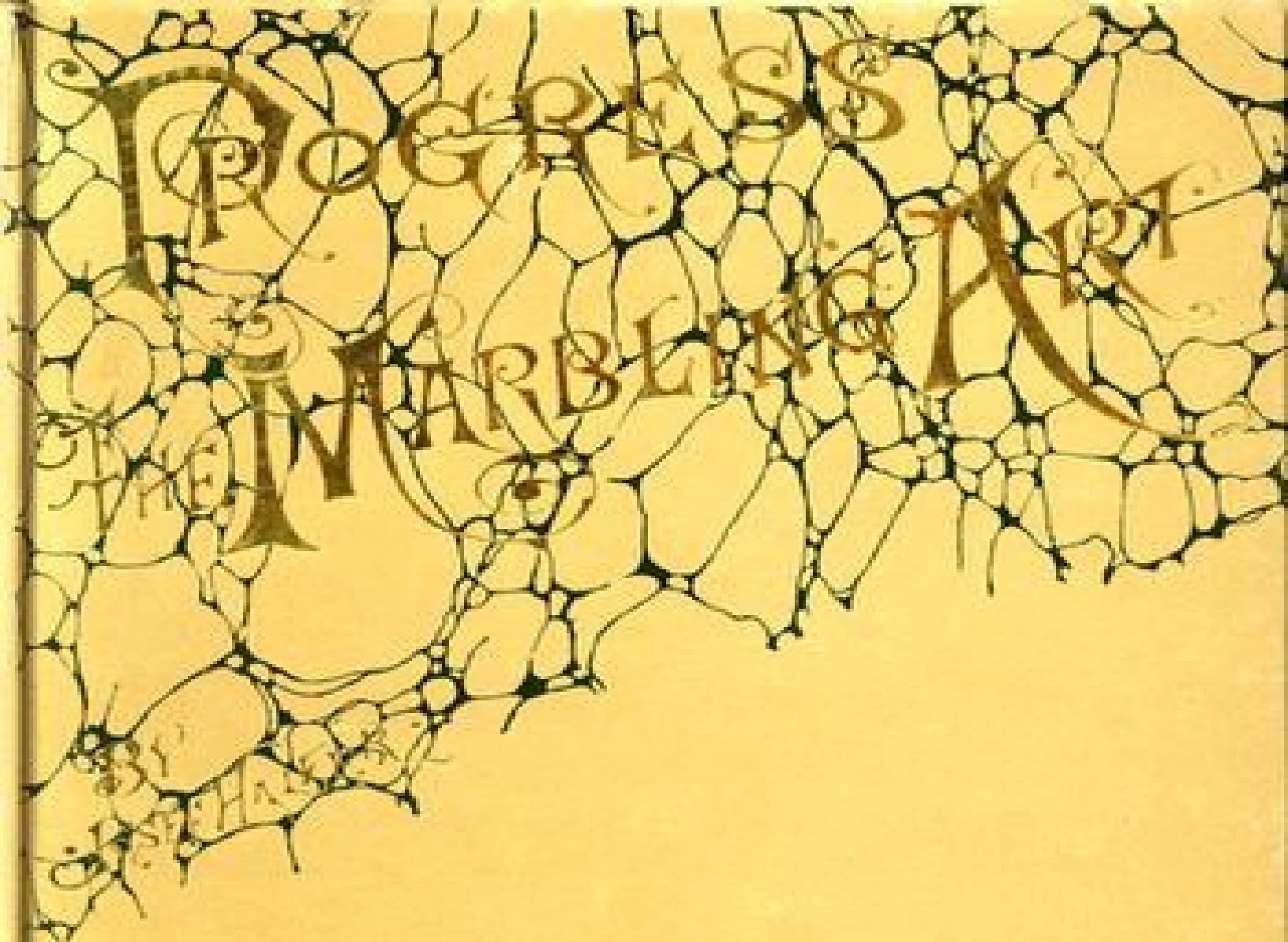
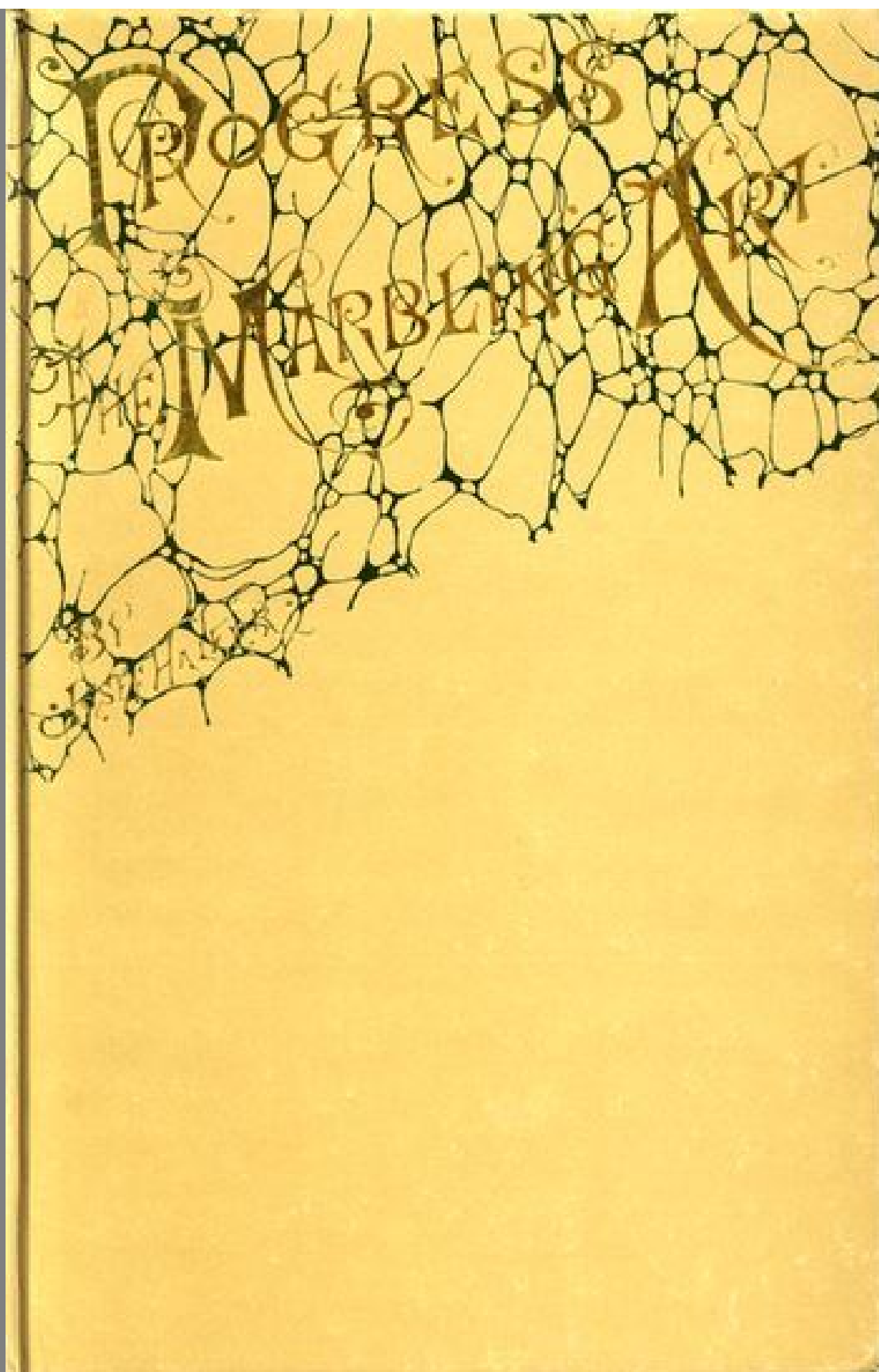






The Project Gutenberg eBook of The progress of the marbling art, from technical scientific principles

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

Title: The progress of the marbling art, from technical scientific principles
with a supplement on the decoration of book edges

Author: Josef Halfer

Translator: Herman Dieck

Release date: October 30, 2012 [eBook #41241]
Most recently updated: October 23, 2024

Language: English

Other information and formats: www.gutenberg.org/ebooks/41241

Credits: Walt Farrell, fh and the Online Distributed Proofreading Team

*** START OF THE PROJECT GUTENBERG EBOOK THE PROGRESS
OF THE MARBLING ART, FROM TECHNICAL SCIENTIFIC
PRINCIPLES ***



Transcriber's Note: The [Table of Contents](#) can be found at the end of the book.

THE PROGRESS OF THE MARBLING ART

FROM
TECHNICAL SCIENTIFIC PRINCIPLES

BY JOSEF HALFER

WITH A SUPPLEMENT ON THE
DECORATION OF BOOK EDGES

Translated by Herman Dieck, Philadelphia

BUFFALO, N. Y.:
THE AMERICAN BOOKBINDER CO.
1894.

Copyright, 1893,
By **LOUIS H. KINDER.**

PREFACE

TO THE FIRST EDITION.

No product finds so many and ample applications as color. Nearly every trade requires it for the decoration of its products so as to adapt them more generally and pleasantly to the present demands.

The art of marbling is that branch of our trade, in which color is brought into use for the decoration of bindings, yet it has not found the desirable general introduction into our book-binderies because practical men have not so intently employed their time and endeavors, to overcome the difficulties, which resist its general application.

Who could solve easier and more correctly than the mechanics, to whom these difficulties and obstacles offer themselves in practice? He is the only one to find the remedy, scientific men not possessing sufficient technical knowledge. Only he is ready and able to stand up for such special trades, to work and to fight for them, who is himself interested and who not only learned to understand the art of marbling from former instructions and traditions, but from his own practical experience.

To him only, will it be possible to gain by close study true points, on which to further develop the whole subject.

My original plan was, not only to remove the technical difficulties of the art of marbling, but also to ascertain the correct colors from the mass at present manufactured, for the purpose of manufacturing marbling colors.

But in this I did not succeed as easily as I expected. Every color manufactory possesses another system to produce its products, and after thousands of experiments I was finally forced to resort to chemistry for the purpose of gaining a knowledge of the raw products viz: earths, metals, acids, bases and their conditions and qualities in order to study their compositions and precipitations, the coloring composite of lake colors and their bodies, and to subject them to investigation on their effects on marbling size.

This was a tiresome work which demanded much patience and pertinacity. During these investigations, I have learned to know the influence of sizing upon colors and how it hindered or expedited a beautiful marbling or comb edging.

Through these experiments I unexpectedly gained a certain knowledge which enabled me to overcome the technical difficulties of the marbling art. The greater the difficulties the more they aroused my zeal to surmount them and the greater my joy when victory was won. During my investigations I saw, that not the color itself, but its body, to which the color is bound either in a natural or chemical way, forms the important part of the colors necessary to our purpose. No branch puts so many critical demands upon the preparation of colors as the art of marbling, because, aside from the most careful grating which can only be done by using the best of painter's colors, there must be a great power of divisibility and excellent covering qualities so that the pigments do not turn pale by the propelling power of the ox-gall expanding on the size, but retain in a dry state, their full lustre. It is therefore impossible without a more thorough knowledge of the chemistry of colors to find from the legion of colors, which are thrown upon the market, the ones best adapted to our art. So much more so as, in most recent times, by simplification in the manufacture of most colors their quality, for marbling, has most essentially deteriorated.

Colors which are used for painting, lithography and printing with the greatest success may be entirely unfit for marbling. The cause is not to be found in their preparation for our purposes, but in the manufacture itself, as the qualities, which are indispensable are not known.

The main attention in the manufacture is directed to the finding of colors free from poison and which will not fade, when exposed to light, if they are intended for a wholesale consumption in painting and the graphical art.

All colors which we intend to use for marbling purposes must have body, because bodiless colors become mixed with the size and run off from the paper in marbling, they therefore can be designated as useless for our purpose. The most important composite parts of colors for marbling are certain substances, among which hydrates of argillaceous earths, tin and lead oxides and sulphate of barium are named as the best.

A great number of coloring matters of organic origin, which are soluble in water, have the propensity in mixing with a solution of salts of lead, tin, and argillaceous earths, with a metallic oxide to form a compound which is soluble with difficulty (and is termed insoluble,) when the latter is precipitated from the solution by an alkali. The precipitations which are obtained in this way are known as lake dyes. Of greater importance for the quality of the marbling colors are the physical conditions of these precipitates, they are either crystallized or amorphous, the former not possessing any value to us.

The serviceable, i. e. the amorphous precipitates, by their quality of great divisibility give excellent colors.

Among the great number of numeral colors there are many which in spite of their amorphous properties are bad marbling colors, such as the compounds of copper, zinc and arsenic. Often it depends upon the manner of manufacturing, even if the color have a good body whether it can be used by us, as, for instance, white lead and barium both of which can be produced as good or useless colors.

The general idea, that light colors are the best, is erroneous, even heavy oxides of metal being excellent colors. Not in weight but in divisibility rests the excellence, for instance I mention pure oxide of iron, cadmium, sulphate of mercury and mercury iodide. These are all heavy metallic oxides and good colors.

Exclusive of the colors the different ways of sizing are most essential parts of marbling. They exercise a great influence on the success of beautiful edges and for that reason I directed my closest attention to them. The more homogeneous and the smaller the viscosity of the mucilage of the sizing, the better for comb, peacock and bouquet edging.

Marbled and veined edges demand more viscosity of the mucilage, which, if the sizing does not possess it, is usually formed during the first stage of decomposition, when the normal time for using it for comb edges has passed. I will mention further on the ox-gall and the sprinkling water, which are indispensable on account of their binding and expanding powers for the purposes of marbling.

It is an easy manipulation to prepare ox-gall, so that it does not decompose, but on the other hand a much greater attention must be paid to the binding and expanding powers of the same, as they may have, in spite of its excellent qualities for color, by insufficient caution in their handling the most deleterious influences on the colors as well, as on the sizing and in most cases are the causes of the failure of the edging. There are several substances, which possess qualities similar to ox-gall and many others which possess only its expansibility.

To the former belong resins which are insoluble but which can be dissolved by the addition of ammonia or borax, to the latter fatty acids of alkali, which we know by the name of soaps and which I shall describe in the article on ox-gall.

Sprinkling-water is mostly used on account of its greater expansibility and its propensity to form circular spots on the marbling ground, (sizing) for veined edges, or as an addition to the ground color of the marble edge. Although it does not possess the binding power of ox-gall for colors in such a degree, it is nevertheless, worthy of our attention. It was the aim of my investigations during many years to bring this all into a harmonious ensemble, and produce a fine and paying work.

In composing this book, I have aimed to arrange it in such a way, that it will be a guide not only to the uninitiated, but also to the practical mechanic. For this reason the book will only give such methods of the marbling as will be crowned with sure success and which have given me satisfactory results.

As books, which are intended for practical men, can only obtain an intrinsic value by combination of facts, gained by practical experience, with scientific principles, I have attempted to attain this object and I hope to have created a lasting work of our trade-literature.

Budapest.

JOSEF HALFER,

BOOKBINDER

May, 1884.

PREFACE

TO THE SECOND EDITION.

It is a pleasure and a satisfaction to every author to write the preface to the second edition of his work, as the sale of the first proves sufficiently, that it has found recognition and commendation.

In the second edition, I have given the results of my uninterrupted studies and the new observations I have made during the last five years. It is my hope not only, that I shall make the study easier to the inexperienced, but also to offer to the experienced marbler a practical compendium in cases of frequently arising difficulties.

The publisher of this new edition has added to it a number of samples, which were collected under my supervision and which certainly will be highly welcome.

I hope and wish that the new edition with its practical arrangement, with its description of my newest observations and with its tables of samples and the other additions and improvements, will find the approval of the trade and will increase the number of friends and patrons of the marbling art.

Budapest,
Very respectfully,
JOSEF HALFER.

October, 1890.

HISTORICAL.



HISTORICAL.

Every branch of industrial art possesses its champions and originators, who have employed their whole intellectual and physical powers to solve problems for the purpose of enhancing and furthering their material welfare.

Often from small beginnings, within a longer or shorter period of time, great enterprises spring into existence, which greatly further our development in culture, science and arts, and are blessings to the whole civilized world. The material welfare arising from them renders it necessary that new champions continuously enter the arena and combat in the cause of progressive industry.

The industries of color and paper making which play most important parts in the art of marbling shall be the first subjects of my essay.

The paper industry which is very highly developed will have a much better future on account of its possibilities in the line of improvement. To-day, in the iron-age, we speak of a paper-age, and for good reasons, because there hardly passes a year which does not bring new discoveries, showing the extraordinary adaptability of paper. One branch of the paper industry which to-day is considered as an especial part of the industry is the manufacture of colored paper, the origin of which, as far as our literature is able to state, extends as far back as the first part of the last century.

From it springs the art of marbling, which in latter years was introduced into our trade.

The manufacture of colored paper makes a second branch of industry indispensable and one which stands on as high a stage of development as the paper industry; this is the manufacture of colors, which deserve the greatest attention on our part on account of the marbling art.

There is hardly a second branch of chemistry which is of such a great old age, historically proven, as the color industry.

We hardly know of a nation on the whole globe which does not make use of color in some way. The use of color for the purpose of embellishment and adornment dates back to historical times. Nature itself, by the beauty of the colors of her flowers and minerals acted as a teacher in the artistic development of the human race.

Egyptian wall paintings show richly developed forms and figures adorned with multi-colored fabrics.

This is a proof that the Egyptians not only understood the art of the manufacture of colors but that they also knew the much higher art of fastening the color on textile fabrics, i. e., the art of dyeing.

In olden times for the coloring of objects, mineral colors which occur in nature the result of a decomposition of metals and earths and had only to be put through a sieve and washed to adapt them to the painters use, were exclusively used.

To alchemy, which was the origin of Chemistry of to-day, we owe a surprisingly large number of artificial mineral colors, for the reason that with predilection it brought metals, earths and mineral compounds within the sphere of its researches which were aimed at the production of gold, but which all were ineffective in this direction. But the time and labor, which were employed in these investigations, were not thrown away. By alchemy an immense number of chemical compounds became known, without which knowledge the chemistry of to-day would not have reached its exalted stage of perfection.

Exclusive of mineral colors several organic colors came into use, the most being applied in the dyeing establishments as they were bodiless colors and at that time the art of binding them to metallic oxides was unknown.

By and by it became known in what way to bind the coloring matter of plants to bodies and to make them serviceable in the art of painting and the graphic arts. Since that time progress in this field has been so rapid that we gaze in astonishment upon the achievements of the color chemistry of to-day.

The progress which developed industrial art and caused it to flourish, was enhanced by a new and highly important invention, the industry of tar

colors.

It is wonderful how man through the vigor of his intellect, tears from nature her hidden treasures which have lain in obscurity for thousands and thousands of years.

Who would have thought, that the antediluvian vegetation with its splendor of thousands of colors should celebrate its resurrection in aniline colors which by their splendid qualities of richness and intensity over-shadowed everything before them? It is a pity that we are only able to use these colors in limited numbers in our trade.

Up to this time they are used more for dyeing purposes, because they are without body and possess two apparently insurmountable proclivities, 1st, they do not mix so intimately with oxides of metal as the organic coloring stuffs of plants and they cannot for a long period withstand the influence of light.

In the meantime it is to be hoped with confidence, that science will soon find the solution for these difficulties. Already the forerunners of body colors have been discovered, a few which could withstand light have followed, as geranium lake in which the coloring matter is only bound to oxide of lead by surface adhesion. Furthermore scarlet lake, which, although bound to oxide of lead, has been put in such a state that it does not run and lastly the artificial krapp-(madder) lake and indigo, which have been produced from coal-tar and nearly equal to the good color lakes of the animal and floral kingdoms.

The main attention in the production of these colors is directed to the end of finding those which are free of poison and will not fade, to make them serviceable to the historic art of painting.

For the aniline color there is therefore a wide field open to experiment.

Whether our interests and demands in regard to colors will be fulfilled is a question of time.

Even the newer productions of the animal and vegetable kingdom already differ in quality from those indispensable to the art of marbling.

I recommend to my honored colleagues to read my book with attention, so that the intended gain shall not be missed and the beautiful art of marbling may attain a larger circle of friends.

THE AUTHOR.

THE SIZE.



THE SIZE.

The first process in marbling is the preparation of THE SIZE on which the colors are to be floated.

By the name of size is meant a consistent glutinous mass, which is obtained by boiling or dissolving with water certain bodies derived from the vegetable kingdom.

Mucilage is found in many plants, especially in the outer skin of seeds, also in many roots, barks, stems and leaves, but its solubility and consistency are very much varied.

Generally the mucilage of plants is classed among the gums; but there is a large difference from them in this, that it will not give a perfect, diaphanous and homogeneous solution, but only swells in it to a thick, massy and viscous fluid, which is similar to starch paste. It is composed of small quantities of gum, starch, sugar and consistent plant-mucilage. If prepared from certain plants, for example the algae of the sea, it also contains small quantities of iodine.

Starch swells in hot water, while with mucilage the same result is achieved by using cold water.

Smaller quantities of mucilage are found in every plant, but larger in gum tragacanth, carrageen or Iceland moss, salep, the leaves of mallows, or of colts foot or tussilage, the roots of marsh mallow, seeds of plantago-psyllum, (flea-bane) linseed and quinces.

Of this number but few are of interest to us, which I shall mention further on.

The density and proper consistency of the size depends on the bodies, from which the viscous or glutinous substance is derived and also from the quantity of water, in which it is dissolved. The consistency of the size is also materially influenced by the temperature, the warmer, the less

substantial and the more pliable; the colder, the more consistent and stiff it is.

A good marbling size should be an entirely homogeneous mucilage, i. e., a solution, which should be free of any vestiges of seeds or strings undissolved. A less viscous size is more adapted to the production of comb-edges, while a more viscous size to the production of marbled or veined edges. The consistency of the mucilage of every kind of size will soon experience an essential change, which becomes apparent by the fact, that the size becomes less consistent, until finally an acid reacting fluid remains, which has no more value for marbling purposes.

This chemical process is produced by the transformation of the sugar parts of the dissolved starch contained in the size into acid and lactic-acid. The more starch and sugar that is contained in the size, the sooner the acid will be generated, while pure plant mucilage containing but little sugar and starch will resist much longer disintegration.

All plant-mucilages, which are soluble in cold water, i. e., swell, will keep much longer than those dissolved in boiling water, because the soluble parts remain inclosed in one cell which is only broken by the boiling heat or by the formation of acid. This is the reason why they only swell up in cold solution.

A homogeneous mucilage will serve for marbling much better, than that, which is produced by swelling, because the latter has an injurious influence on the fine distribution of color for veined, comb and other edges. Although the surface of such a size seems to be smooth, the eye will discover on closer investigation the undissolved bodies in the small granules of the size.

The mucilages obtained by swelling are always milky-cloudy and never permit the colors distributed upon them to expand with the same purity and smoothness as the homogeneous varieties, on account of the small granules hindering the drawing of the colors on their surface. On the other hand there are varieties of size, which are very viscous on account of a too homogeneous mucilage and therefore from their nature are not adaptable to any kinds of edging which must be drawn by the stylus. The colors can simply not be drawn on the surface, because this very viscous mucilage is drawn along by the stylus instead of being cut by it.

In investigating the different varieties of size I have noticed a very surprising action of the extremely viscous mucilage, which nearly confused me for the moment. I have scalded with boiling water plantago-psyllium (flea-bane) and have beaten it with wicker-rods tied together so as to produce in a better manner, the mucilage found on the outside of the seed. After cooling it off, I filtered it through a linen cloth and poured it into the trough but have not waited till the very viscous thread of mucilage parted after the trough was filled to a certain height, but have placed the vessel on the floor and to my utter surprise the thread of the size spread out and reached the vessel on the floor, becoming gradually thicker until finally the whole size flowed in a white stream with great velocity over the rim of the trough (2 inches high) and returned to the vessel standing on the floor until the trough emptied. This was done in a very short time. The cause of my surprise was in the very consistent mucilage which took along ever-increasing quantities of the size and finally entirely emptied the trough.

Prepared colors, which on a less viscous size are normally distributed expand with a greater velocity to a greater extent on a more viscous size and therefore become pale.

For such kinds of sizes the earth colors are preferable but only for marbled edges.

The normal strength of size can be discerned without araometer or other recommended apparatus. My opinion is that $\frac{2}{5}$ of an ounce of good carrageen moss to a quart of water are sufficient, as carrageen excels all other glutinous bodies by the quantity of mucilage contained in it. To test the correct strength of the size pour some of it into a flat vessel, as a plate or dish, and throw a drop of color upon it; if the color does not spread out, but rather sinks down a drop of ox-gall must be added. Renew the test and so on until the color spreads out to about 2 inches, then take the stylus and draw the color into lines as in producing comb-edges. If the size be too thick the color will be drawn along by the stylus and cannot be cut clean through. On the other hand if the size be too thin it can be seen immediately in its quivering and the running of the color, which cannot be drawn into straight lines. In the former case some clean water must be added to the size, in the latter case which, if the size be well boiled, seldom occurs, thick size must be added by boiling more. The real consistency of the size is so

easily learned in practice that a mere feeling with the fingers will be sufficient.

The preparation of the colors is carried on separately on a small part of the size, so that the whole quantity is not rendered useless. If the colors are rightly prepared in proportion to their expansibilities in relation to each other, the new size is to be poured into the trough and the marbling may begin.

The mistake which is commonly made in marbling is, that too little attention is paid to the size and color. Frequently the size is soiled and the colors are spoiled by adding too much gall before an edge is produced. For this there are generally three causes, either the size is too fresh, too thick or the colors have been prepared on the same size by which the edges are to be produced.

Each size which is boiled should only be taken into use after perfect cooling and after several hours have elapsed, since in a warm state it would not permit the colors to expand even by adding a large quantity of gall. Only after the expiration of ten or twelve hours after boiling will it have the correct quantity of mucilage and then only the size possesses its full value. This time having elapsed, the size is filtered through a cloth and is then ready for use.

A size when too thick, allows the colors to expand only with great difficulty and demands a three fold larger quantity of gall than when it possesses its normal consistency. It is very difficult to remove from a too consistent size the remnants of colors.

They always unite with the size and in this way the size not only becomes soiled but it is impregnated with a large quantity of gall. As I said before, the colors should always be prepared on a separate size which can be poured into a flat vessel as it is nearly impossible to prevent several drops of color sinking to the bottom.

If this precaution be taken and if the colors are prepared according to their relative expansibilities as they are used in the order prescribed, the process will not meet with any obstruction or difficulty. As the size plays such an important part in marbling, great attention must be paid to its preparation if

it is desired to make a good base for the colors. I especially recommend to laymen that they direct their attention to the consistency of the size, as this is always the key to the production of good edges.

The different peculiarities of the varieties of sizes, viz; evaporation, influence of temperature, starry formation of colors, formation of angular scales in comb-edges, and all other things which have influence on colors, will be treated of in a special article on the varieties of size.

CARRAGEEN MOSS.



CARRAGEEN MOSS.

The Carrageen or Iceland moss, or lichen is one of the algae of the sea. *Chondrus Crispus* grows on the cliffs of the Atlantic Ocean and the North Sea. Carrageen smells slightly of iodine and when uncleaned tastes offensively salty. Like all seaweeds it contains a great quantity of mucilage and small quantities of salts of iodine and bromine. When washed in sweetened water it becomes nearly tasteless and is without smell. On account of the great quantity of mucilage contained in it, and of its action as an alleviative to pulmonary complaints it has found many applications in medicinal practice and is given to the patient boiled in water, milk or bouillon or as gelatine in cases of coughs and diarrhoea, being a slightly nutritious substance.

The carrageen moss or Iceland moss or weed is a lichen and consists of a tuft of flat deeply divided cartilaginous fronds of an olive brown, sometimes yellowish or reddish white color; the tuft is sometimes 2 to 5 inches long.

The moss is used in many ways; as size for marbling edges, for clarifying beer, as linen-weavers starch, for finishing cloth and in the production of iodine.

The good pure carrageen is horn-like, yellowish-white, contains a great quantity of mucilage and gives the best size for marbling, as it complies with all the requirements of a good size. It is homogeneous, of great consistency, not extraordinarily viscous and cheaper than all the other materials containing mucilage which are used for marbling, but is inferior to gum tragacanth as far as durability is concerned, and that it must be subjected to boiling, but this trouble is amply repaid by the beauty of the productions.

On this size all varieties of edges can be produced, and by its diaphanous watery mucilage it gives an excellent ground to work upon.

The preparation of carrageen size is confined to one ever recurring manipulation, which must be repeated every eight days, if an always good and useful size is desired; the expenses are very trifling, a few cents for the best kind of carrageen. If bought in larger quantities the expenses for carrageen can still be reduced. For the preparation of the size take a new pot, which must always be used for this purpose exclusively, pour in six quarts of water, which quantity is sufficient to twice or three times fill the usual marbling trough, add 2-1/2 ounces of Carrageen moss and boil until the water bubbles up several times. A longer boiling is injurious because the mucilage becomes flaky and muddy. 2/5 of an ounce of Carrageen to one quart of water is fully sufficient to produce a consistent size, provided that only the best Carrageen is used.

Soft water or rain water was formerly used exclusively for preparing the size, as by using it a greater durability was obtained.

During my researches, regarding the durability of the varieties of sizes, I found means by which the influence of hard water on the ground can be suspended, so that spring and pump water can now be used without any danger.

Every kind of hard water contains mineral salts, spring water especially containing lime in smaller or larger quantities. This can be seen, when the soap, which is used in such water, curdles, because of the ingredients of soap stearine palmitic or fatty acids of sodium or potassium, form a combination with the lime, to carbonate of lime, and the potash soap is converted into an insoluble lime soap. If we add to hard water a quantity of sodium or potassium, the carbonate of lime is precipitated as a white or yellowish powder, according to the quantity of iron contained in the water, and a soft water remains.

The sun produces on a great scale in nature, what the chemist achieves on a smaller scale by evaporation or precipitation and cooling of steam, a water free of all mineral salts, which we call distilled water. Rain water is the same, but the flowing water derived from a river already contains mineral salts in greater or smaller quantities, which are brought into it by different springs and substances. The water gained by precipitations is considered soft. Carbonates of sodium or potassium are strong bases, exactly the

opposite of acids, and they therefore prohibit the oxidation in plant mucilages. This chemical reaction against the formation of acids I used for the purpose of prohibiting a deterioration of the size by adding sodium, in which I was thoroughly successful, as it can easily be preserved in a cool place for even eight or ten days.

Two obstacles offered themselves, which I have overcome after a tireless and thorough investigation, first the strongly alkaline property of the size which was generated by the added carbonate of sodium and second the fermentation of lactic acid which occurred notwithstanding the sodium and made the size useless.

The presence of alkalies was injurious to such colors, chemically bound to bodies, among which red is most affected. The coloring matter was removed from the bodies and became mixed by running with the size, causing the size to become soiled. Even the shade of the color was changed to another which was nearer to purple, and in this way, the beauty of the color was greatly lessened. The second obstacle is the fermentation of lactic acids which always occurs where starch flour is in the solution, causing in consistent sizes a gradual thinning out and even a destruction of the size.

It was not my aim to save the size from destruction forever, as this is impossible (laws of nature cannot be obliterated at will) but to keep it useful for a certain period of time, and this I achieved after many experiments by using borax.

Borax is a neutral salt of strongly antiseptic properties, prohibiting fermentation and decomposition where neither base nor acid predominate. Borax is produced by neutralization of boracic acid with carbonate of sodium and can be bought at every drug-store.

This salt is sold in crystals of the size of walnuts; for our purpose it is ground to powder and is kept for use in a closed vessel. The size is first used without any addition of borax as it is then better adapted for producing comb and other drawn edges, and only after some time has elapsed, in summer 36, in winter 48 hours, are 150 grains of borax added to the size, whereupon the size, having been repeatedly stirred to dissolve the salt, will keep entirely good and useful for a period of eight days without doing any harm to the colors. On a size which contains borax, the colors expand in a

much greater degree than upon unadulterated size, without impairing the consistency of the mucilage. An unadulterated size is therefore, better for marbled and veined edges, but if it be desired to produce only marbled and veined edges on the size, it is advisable to add borax to the water before boiling the carrageen moss.

As the influence of temperature is important in marbling, and the production of beautiful edges is often jeopardized by it, it should therefore have the most careful attention. Every glutinous mass acquires, according to the difference of the temperature of the size and air, in one-half to one-quarter of a minute, a film (top) which forms through natural evaporation.

On a size which has such a film, or top, the colors do not spread out in such circular forms as on a size on which the colors are quickly thrown right after the film (top) has been taken off, but form star-like veined spots which are torn by the film in all directions.

Even in the normal expansion of the drops of color, it often happens, that the rim of the drop is jagged, the cause being that the size already possessed a thin top, which was not yet sufficient to prevent the expanding of the colors. If the size be left standing for a half-hour or an hour before throwing on the color without having at first removed the top of the size, the drop will tear a hole into this film and will sink, even if quite a large quantity of gall had been added. This explains the contraction and the sinking down of colors on a still warm size, inasmuch as the top is formed by evaporation of the warm mucilage, so quickly and so thick, that the color which had already begun to spread out is pressed together by it and is forced to sink.

The greater the difference of temperature between the size and air, the quicker is the top formed. It is therefore a rule, that the size should be allowed to cool off in the same room, in which the marbling occurs, because in this way a symmetrical degree of heat or cold of the size and air will be gained. On such a size, the formation of the top is much slower and thinner, so that even after one to one and a half minutes after removal of the top of the size, the colors can be thrown on without the occurrence of the starry formation.

I will here mention a fact to show the importance of temperature. As this point has given me much trouble and has taken much time and labor, it was

my aim to find that mysterious effect which I was unable to overcome despite my numerous chemical experiments, until my eyes became opened and I saw accidentally that this obstacle was the difference of temperature of size and air. From that time on I was able to prosecute my labors without the slightest difficulty.

To show the excellence of my marbling colors, I used a great number of samples, perhaps several thousand, and in order to produce them I worked on Sunday so that I could employ my whole time to the good purpose. It was winter; on every Saturday I ordered two pots of size to be boiled, and to cool them off I placed them in a room, in which there was no heat. On Sunday morning I filtered the size through a loose linen cloth; it was as clear as crystal, it possessed the necessary consistency, and I was much pleased with the results, which I had expected to achieve.

I commenced the preparation of the colors in a separate flat vessel, using but little size, so as to keep the rest from being soiled, but I was greatly astonished, when the colors in spite of the addition of gall, which of course was carefully done by drops, first spread out and then again were contracted, the starry formation of the colors took place, whenever I was not quick enough after the taking off of the top to throw on the colors.

I was dumbfounded and could give no explanation for it; the size was sufficiently cooled off, it had been standing the proper length of time, the mucilage was excellent but in spite of all that, these difficulties occurred.

Thinking that I might be more successful, if I should do the marbling in the trough where there is more surface offered to the color to spread out, I filled the trough with size of a sufficient height and then commenced the marbling, but immediately after the first trial I instantly recognized that even here my experiments would have the same result. Although I threw on the color with great alacrity, right after the top had been taken off, the first drop spread out normally 4 inches, the second but three, the third two, and the fifth and sixth only one inch. In this way I received instead of an equally wide band of thrown on color, one which tapered to a point. If I had not worked according to the system which I shall describe in the chapter of throwing on of colors, I would have received already with the fourth drop, a star-like formation.

Whatever I did to avoid this obstacle was worthless, it was and remained a mystery and I was unable on that day to produce anything beautiful or worthy of consideration. This happened several times until, by accident, and through careful study I found the cause which was nothing but the fact, that the size was much colder than the air in the room in which I worked. The evaporation on the surface of the size was so great, that the top was formed nearly as quick as on warm size and therefore a favorable result was impossible.

Another abnormality which arises from cold temperature acting on the size is the fig-like formation occurring during the drawing of the colors. The cold in some way seasons the surface; this surface becomes mixed in filtering with the liquid part of the size in numberless small particles which although the size is well stirred up in the trough by the fingers, are not dissolved and this hinders the drawing of the colors.

They adhere to the stylus and follow it and arrange themselves between the color lines in a way similar to a green fig, having a broad upper and a pointed lower end. Such an edge traversed by the comb will give an ugly result. A difference of temperature of the size and the air can also occur when the working room is suddenly heated, or, when in summer time, the size is kept in a cold cellar. To effectively obviate this difficulty it is necessary to bring the temperatures of the air and size carefully to the same height which is made easy by leaving the size, after boiling, in the room, where the marbling is executed.

In winter, when the room is heated, the size should be poured into the trough one hour before using, in summer the thing regulates itself because it is then unnecessary to keep the size in a cool place, the addition of borax being a sufficient protection against spoiling. But in spite of it all, in winter-time when in drawing the colors the fig-like formation should occur, it is advisable to return the size to the pot and to warm it up slightly so that the seasoned particles are dissolved, until an equal temperature is reached, whereupon after the lapse of one hour the marbling can again be proceeded with.

Should the size be too thick the trouble can be remedied by the addition of some warm water.

The quantity of mucilage contained in carrageen moss varies greatly, and the consistency of the size often depends upon the quality of the moss. The quantity which I gave as essential to the preparation of the size means first class moss.

Should the consistency of the size be insufficient after the first trial then from 1/2 to 1 ounce of moss should be added as it is clear that the carrageen moss is not of prime quality. Before I speak of another variety of size, I will repeat the main condition which should meet with the attention of every layman and which are as follows; the size must not boil too long, in order that it may not become flaky and muddy, it should remain for cooling in the same room in which the marbling is carried on, so that the temperature of it and the air are the same; the colors should be prepared separately on a small quantity of size so that the whole size cannot become soiled and, lastly, the top of the size should be taken off before the colors are thrown on, so that no film can form before the colors cover the surface of the size.

The quicker the manipulation of marbling is executed the prettier are the edges produced. As soon as the size is covered with color, the formation of the film is prevented.

As the size is one of the most important parts in the process of marbling, I recommend these main points to the special consideration of the reader.

GUM TRAGACANTH.



GUM TRAGACANTH.

Gum tragacanth, also called gum-dragon, is the product of various species of astragalus which is found in Greece and Turkey and is a natural exudation of shrubs and from exudations resulting from incisions made in the stem near the root. This shrub is from 28 to 35 inches high and two inches in diameter. It is collected in July and August after the exudations have ceased. It is a hard, tough substance more or less white according to its purity in very irregular flattened shapes and in tortuous vermicular filaments. It contains aside from a little gum and starch, a consistent plant-mucilage without smell or taste and it is used in technical ways, in printing of textile fabrics and finishing, in producing plastic masses, as a binding medium for the production of confections and in the book-bindery for marbling size.

The leaf tragacanth of Smyrna is the best. It is brought to the markets of the world via Constantinople or via Smyrna in boxes of 100 to 200 lbs.

It forms flat ribbon-like pieces which consist of peculiar layers of horn-like consistency, white and diaphanous; when broken it is dull and splintered.

An inferior quality is the tragacanth of Morea, which generally comes from Greece via Trieste. This consists of oddly shaped, peculiarly twisted pieces, partly of pure white, partly of yellowish and brownish colors. For a long while, tragacanth was known as a good material for marbling size and, for that reason, it is used to-day in a good many of our book-binderies for this purpose.

It is much more liked than carrageen moss because the consistency of its mucilage and its durability are great and the preparation of the colors does not demand such great attention. But since my experiments and investigations upon the excellent effects of borax on the durability of carrageen size have become known, tragacanth has lost much of its popularity, because with it the edges can never be produced in a similar fine way as with carrageen size and besides the price is higher. Tragacanth is one of these short-viscous plant mucilages which swell in cold water but do

not give a perfectly homogeneous solution. The mucilage consists of innumerable small granules, in which the starch is enclosed by cells.

These small granules prohibit, within the first two or three days of a fresh tragacanth size, the drawing of edges, as they make the colors thrown on appear rugged, and in drawing injure the fine hair lines. Five or six days after the dissolving of the tragacanth, the mucilage becomes more homogeneous and therefore better adapted for marbling.

If the swelled up mucilage is boiled, after the first or second day, then the solution will become perfectly homogeneous and will be as good as carrageen size except that the colors, which normally spread out on carrageen size, will expand much more on tragacanth size as it possesses more consistency and therefore they will become paler. The more dense the size, the thicker colors and the less gall is necessary to produce a normal expansion. The same difference in the power of expansion of the colors as in carrageen and the tragacanth is noticed in colors which normally spread out on tragacanth but which, very largely, expand on the size of plantago-psyllium (flea-bane) because the latter has the most consistency and homogeneity of them all. The different effects of the varieties of size on the colors are due to the consistency and to the different conditions of gravity. The objections of the consumers of my marbling colors who use tragacanth, are, that they consider the edges too pale. The cause of this is that my products are only prepared for carrageen size, inasmuch as my investigations have shown that it is the best, the cheapest and the most adapted for all varieties of edges.

The homogeneousness of the tragacanth mucilage in cold solution comes naturally after the size is five or six days old, because within this time fermentation of lactic acid occurs, which opens the cells of the small granules of starch while at the same time boiling heat has an immediate result in the same direction.

As soon as these cells are opened the formation of acids of the sugary parts and the fermentation of lactic acid of the starch contained in the size take place as fast as in any other variety of size, hence the size of tragacanth has no superiority over any other, besides the best, or picked-leaf tragacanth of Smyrna costs about four times the price of the best Carrageen.

For the preparation of the size, take 3 ounces of tragacanth, pour two quarts of water over it, leave it stand for 24 hours, then stir well and leave it standing for 12 hours more, repeat this until the homogeneous thick mucilage has been produced, then add 4 quarts of water, again stir it up well and filter it and the size is ready for the marbling process.

For tragacanth size, colors of great consistency, mostly fine earth colors are the best. They must be ground exceedingly well and very little ox-gall is to be added. But as these earth colors lack in the power to spread out and in divisibility, a characteristic of colors prepared for carrageen size, and as they never will have such fineness and smoothness, always appearing rugged, it is impossible for me to recommend gum tragacanth for the preparation of size.

I must mention further an effect, which earth colors exert on tragacanth size, viz., that they can be used on paper not prepared with alum, without running, while this is not the case with colors, which were prepared for the Carrageen size. In another chapter upon ox-gall I shall explain why the colors used with Carrageen size must be transferred on alum paper.

Salep, Plantago-Psyllium.



SALEP, PLANTAGO-PSYLLIUM (Flea-bane) AND THE OTHER GELATINOUS BODIES.

Radix salep are called the dried tubers of several species of orchids, they are round, quite flat, yellowish white, horn-like, semi-diaphanous, very hard and without taste or smell. Formerly salep was imported mostly from Persia, but now the tubers of orchids grown at home are collected and do not in any way differ from those coming from Asia. They contain a good quantity of gum and on that account are used for finishing silks and for medicinal purposes. It is sold ground, or as powder and can be bought at every drug-store. As marbling size, salep possesses very excellent properties, similar to carrageen moss, but its high price prohibits its general application. In preparing size, use, to 6 quarts of water 2-1/2 ounces of powdered salep which must be well boiled for a long period so that it may become entirely dissolved. After standing 24 hours, it can be used in the same way, as carrageen size.

Plantago-Psyllium (Flea-bane, Flohsamen, Flohkrautsamen) is the seed of different species of Way Bread belonging to the family of the plantaginea viz., plantago-psyllum, Pl-Arenaria, Pl-Cynops, the second of which is found in Eastern Germany, Switzerland, Hungary, etc. on sandy fields, the other two on the sea-shores of southern Europe.

The seed is small with one side flat and the other concave and has a furrow upon which is located the eye or navel. It is lustrous and of a reddish brown color, the best varieties coming from southern France. In consequence of the great quantity of plant-mucilage or gum, that it contains, by treating with hot water a consistent mucilage or size is obtained and used in the manufacture of marbled paper and for finishing textile fabrics.

To prepare the size, place 4 ounces of flea-bane in a wide wooden tub, pour 6 quarts of boiling water over it and beat well with a switch of wicker-rods until the gum is entirely separated from the seeds, allow it to cool and after 10 or 12 hours filter through a linen cloth, when it is ready for use. The mucilage of flea-bane is quite homogeneous and of great consistency, is

very viscous, forming strings very readily and is therefore not used for drawn edges since the colors will be drawn along by the stylus without being cut. It is mostly used for producing thickly veined, ordinary marbled edges, but flea-bane size is also much more expensive than that of carrageen moss and on the other hand its quality is so much inferior to that of the other that I cannot recommend it for use as size.

I will briefly mention the other vegetable matters containing an especially large quantity of gum or mucilage, but which have not come into use for preparing marbling size on account of their cost and of the less consistency of their mucilages.

The richest in this regard are the seeds of quinces and linseed. Their mucilage is as viscous as that of flea-bane. Much less gum is contained in the leaves of mallows of colts-foot or of marsh-mallows.

In closing this chapter upon the mucilages of plants and their applications as sizing for marbling I again recommend carrageen moss as the best size, because it has given me during my investigations and in practice, surprising results of its usefulness for all kinds of marbled edges.

OX-GALL.



OX-GALL.

THE CHEMICAL CONSTITUENTS OF OX-GALL.

In the bile of every animal are two peculiar acids both containing nitrogen. One of them free of sulphur, the other containing it, both by boiling with acids and alkalis yield the same acid free of nitrogen, which is called cholic acid.

If these two acids be separated from each other in a chemical way we obtain glycocholic acid, an acid free of sulphur and the other taurocholic acid containing sulphur. The alkali salts contained in bile are quite soluble in water or alcohol but insoluble in ether. To produce it in a pure state mix the bile with as much animal coal, (spodium) so that finally, on evaporation in the water-bath a dry powder results, from which is drawn, by the use of absolute alcohol, the now colorless cholic acid esters, cholesterine, choline, lactic acid, etc. When ether containing a little water is added, the cholic acid esters are precipitated as colorless plaster-like masses and crystallize, when left in the fluid for some time.

The so-called crystallized bile is the final material for the preparation of acids. The acids of ox-gall have been investigated in the most thorough manner.

Glycocholic acid is obtained as a colorless precipitate from a watery solution of crystallized ox-gall, by adding dilute sulphuric acid until the fluid becomes flocculent, which after a while, especially, if the water contains a little ether, is changed into a voluminous 'magma' of fine white needles.

If collected upon a filter, these needles unite and form a lustrous felt.

Glycocholic acid is sparingly soluble in water but soluble in alcohol. It forms the principal constituent of bile, it reacts feebly with acid and tastes bitter-sweet. Taurocholic acid is found only in small quantities in ox-gall and has no important effect on the colors.

THE EFFECT OF OX-GALL UPON COLORS.

By chemical analysis we have seen that glycocholic acid is the principal constituent of ox-gall and in combination with taurocholic acid, glycin, choline and lactic acid is soluble in water. On the other hand, when pure, it is soluble with difficulty and therefore is only of use to us when in combination with other acids, as it must combine intimately with water-colors.

The atomic weight of gall and the insolubility of pure glycocholic acid in water are the most valuable properties of it for the marbling colors.

The first gives to the colors the expanding power, the latter the adhesion to paper.

All the lower fatty acids, among which ox-gall may be numbered, possess properties similar to fatty bodies, namely a less atomic weight than water. They differ from fatty bodies in this, that they mix with water.

The expanding power and lightness of ox-gall are, consequently, peculiarities of equal value. Marbling size forms the base for the colors, which by the consistency of the glutinous mass and by the gall added to them are prevented from sinking to the bottom. They float like drops of oil upon the water and only differ from them by the fact that the colors will spread out on the size, while the drops of oil on the water are contracted to circular spots of small size. If a drop of oil were thrown upon the size it would spread out very quickly.

This purely physical occurrence is caused by the different weights of the bodies and the consistency of the size. The fatty bodies, which are much lighter than water, float upon it but the gravity of the drop itself prohibits its spreading out, as it sinks somewhat and is held together by it. It is different with the size; by the consistency of the glutinous mass the gravity of the drop is antagonized and as a result it flattens and spreads out.

As I have already mentioned the insolubility of glycocholic acid binds the color to the paper. To explain this I will mention a few examples; the saturated alum solution will cause glycocholic acid to curdle. If, therefore, we add to ox-gall which is basic the saturated solution of alum, a soft

plaster-like mass is precipitated which consists of glycocholate of aluminium, insoluble in water. This chemical reaction explains the durability of the colors. If we desire to marble books or papers we impregnate them with alum water, allow them to dry partly and take the edge from the size.

In the same moment that the paper soaked with alum comes in contact with the colors, the latter become bound by the formation of glycocholates of aluminium and do not run. They have the same property of fixing colors that resinous soaps used in the paper manufacture for gluing machine paper have. If this resinous soap were not too strongly basic, which is injurious to many colors, it might be applied as a surrogate for ox-gall.

As the resinous soap on account of its binding quality and insolubility could also be used in marbling with such colors, upon which it exercises no injurious effect, I will here give its preparation and former use in the manufacture of paper. If 2-5 ounces of rosin be boiled with 1-1/3 ounces of sodium lye and if you add by teaspoonful from time to time so much lye until a sample of it is dissolved by hot water to a clear fluid, the mass on cooling congeals and forms a quite solid soap. If we mix a solution of resinous soap with one of alum then we obtain an insoluble compound of resinous acids and aluminium. In this way resinous soap is used in gluing paper by adding to the paper pulp, first resinous soap and then a solution of alum. During this process a thin layer of insoluble soap of aluminium is formed around every particle of the paper and thus the running of the ink is prohibited. The same process occurs with ox-gall contained in the colors, when the edge is lifted off on paper or book-edges impregnated with alum, by which the running of the colors is prohibited.

THE PREPARATION OF OX-GALL AND ITS USES.

The preparation of ox-gall to be used in marbling is simple. Take a quart of fresh bile (ox-gall or fish-gall), place in a bottle which contains when filled from 1-1/2 to 2 quarts, add 1/2 pint of absolute alcohol, shake well and leave stand for from 14 days to 3 weeks. Within this time all particles of gum and all fatty substances which are present in some galls in comparatively large quantities, will fall to the bottom and the gall will be

thin-fluid, pure and diaphanous, brownish, yellow or greenish according to the nourishment of the animal from which it was taken.

Cow-gall contains more gum and fatty substances, than ox-gall; fish-gall, on the other hand, is thinner than ox-gall and would be the most useful of them all, if it could be obtained in sufficient quantities. If the gall is thin and pure it is filtered through paper which is easily and quickly done as it runs like water. If by filtering the dregs of the gall the filter should become clogged, a new one must be used.

The addition of alcohol causes the precipitation of glutinous and fatty substances and preserves the gall from decomposition. If prepared in such a way it can be preserved for years without spoiling.

All colors which we intend to use for marbling must be bound to bodies absolutely insoluble in water; it is therefore a mistake to say, that the colors dissolve in water. You may grind the color on a marble slab or in a machine as long as you please, but you will only obtain a great degree of fineness of the bodies but never a dissolution of the same.

Each body possesses its limit of divisibility; in amorphous bodies the high divisibility is but natural, but in crystalline bodies this division must be produced by grinding or washing.

The gall does not combine with these bodies, nor does it penetrate them, but only clings to them loosely. It can be readily removed in case the colors should be rendered useless by the addition of too much gall. The color is allowed to precipitate in the bottle and the water standing above is poured off, fresh water is added and this manipulation is repeated several times.

The gall which surrounds every particle of color forms, as it were, the support of it and adapts it to float upon the size.

Bodiless colors, which give a complete solution with water will run into each other on being thrown on the size and will flow from the paper when it is lifted off. The insolubility of the color bodies therefore prevents them from running although they are disarranged on the size in drawing and although one color may be compressed or expanded by a second, yet they all remain separate without mixing, except, perhaps, that the shade of the

first color becomes more intense, because its color particles are pushed together by the more violent expansion of the second color.

From this it will be seen that the colors, to be useful for our purposes, must be thoroughly insoluble. The gall is added during the process of grinding the color, so that the particles of colors are fully surrounded by the gall. The gall has an excellent effect on the colors but it also can act very injuriously if the necessary precautions are not taken. Carelessness is mostly the reason that the edges do not possess the demanded lustre of color and why they appear pale, as the marbler often uses the gall too soon when he notices the smallest obstacle, (due in most cases to the size.) It is therefore not astonishing that brilliant comb or peacock-edges are so rarely seen.

It is an obstacle to marbling, that the gall mixes so easily with the size. It often happens that the gall spoils the size before an edge was ever produced on it. This happens especially when the size on which the colors are prepared according to the old method, is too thick. The size is frequently soiled and spoiled when the colors are prepared, because the colors can not be perfectly drawn off on thick size. There will always remain some particles which will not only soil the size but impregnate it with gall, and which will cause the entire uselessness of size and color.

A very consistent size will make the preparation of colors extremely difficult, as they need a double quantity of gall for the purpose of spreading out. If there is but one color used, the preparation on such a size would be possible without spoiling it, but with four colors this is entirely impossible because the repeated drawing off of the colors, which always leaves particles behind, will, by and by, impregnate the size so that when the fourth color is prepared the first will not spread out any further.

The more the impregnation of the gall and size increases, the power of expansion of the colors decreases and this continues until both materials are useless.

It is therefore advisable, as I have already stated, in the chapter upon the varieties of sizes that the colors should be prepared separately on a small part of size to determine the correct consistency of the latter and to prohibit the whole size being soiled.

The gall should be kept in a small bottle containing about 1/10 quart with a perforated stopper from which a small tube protrudes and from which the gall can be added to the colors in drops.

Although the preparation of the color in this way takes more time, this trouble is amply repaid by the result.

Fatty bodies are injurious to the size, therefore they must be carefully avoided because they have the same effect as the gall, they form, although not insoluble in water, a combination with the size and prohibit the colors from spreading out. Fatty bodies can be transferred by glutinous fluids into a state of the most minute division and they then form emulsions.

Natural emulsions are milk, the yolk of egg, and the milky saps of plants. For this reason, in many establishments raw milk is used as a propelling medium for hair-veined edges.

There are also fatty bodies which, in an artificial way, form an emulsion even with water; for instance, almond, poppy and hemp, if they are ground to a pulp with a little water yield a milky mixture. All these emulsions artificial as well as natural, can be employed as expanding mediums and give better results for marbling than petroleum or naphtha.

Substances Acting Similarly to Gall.

SUBSTANCES ACTING SIMILARLY TO GALL.

There are rosins which are soluble in ammonia or borax. These solutions possess properties similar to gall and either can be used as an expanding medium or as an addition to colors. Different experiments with these solutions gave very good results.

To produce them pour a quart of water into a vessel, warm, add $\frac{2}{5}$ of an ounce of shellac or rosin and a $\frac{1}{100}$ part of a quart of ammonia or $\frac{2}{5}$ of an ounce of borax so that the rosin becomes dissolved, and then bring the mixture to a boiling temperature.

Ammonia dissolves the rosins much more quickly and thoroughly than borax, but the solution in ammonia has the property of gelatinizing the colors after they have remained standing a short time. They therefore can only be used in a greatly diluted condition. The borax solution, on the other hand, has no such effect; not even the concentrated solution. I mention these two solutions especially, to instigate further investigations.

SPRINKLING WATER.



SPRINKLING WATER.

Sprinkling water is one of the newer discoveries in the art of marbling and deserves full consideration on account of its good properties. Its advantages consist in this, that it produces circular forms upon the size and has a greater expanding power than gall.

Sprinkling water is used for hair-veined edges and as an addition to the ground color in marbling.

To prepare it take 1-1/2 ounces of Venetian soap, finely chipped, place them in a pot with a quart of alcohol, warm up some water in a second and large vessel in which the pot containing the alcohol and soap is put, bring the whole to boiling, when the soap will be entirely dissolved in the alcohol. Keep a quintuple quantity of water in readiness into which the solution of soap is poured and stir well, the sprinkling water is then ready for use. The solution of soap in alcohol is carried out in a water-bath to avoid the ignition of the volatile vapors which are generated by this process. The alcohol prevents the foaming of the soap, the soft water the curdling.

The solution when ready is to be kept in a closed bottle.

Generally to the colors which are used for hair-veined edges and marble more gall is added than to those used for drawn edges. It is, therefore, easily understood that with these, a more intensively acting expanding medium must be used to spread out the strong colors into veins.

The sprinkling water is here of great service, because, although ox-gall in its pure state possesses a sufficient expanding power it is here more desirable to produce beauty of forms. In this respect the gall is much inferior to sprinkling water, the former producing all kinds of forms, the latter only beautifully round and oval ones.

For marbling, the ground color is prepared with gall so far, until it slightly pushes the other colors aside and then, by and by, so much sprinkling water is added until the desired expanding power is reached.

Every ground color which is prepared with sprinkling water spoils on standing two or three days becoming slimy and viscous, and, for economy's sake, not more color should be prepared with sprinkling water than is required for one marbling process.

For hair-veined edges, the sprinkling water is used in the same way as gall. Use a large painter's brush or whisk and move the colors by beating into veins, with this brush or whisk through a slot, or throw the sprinkling-water upon the color by means of the sprinkling brush and sieve. In both cases the same result will be achieved.

ALUM WATER.



ALUM WATER.

In the chapter upon gall I have mentioned in a general way the action of alum water upon the edges, it remains only therefore to explain the preparation and application of the same.

To achieve beautiful results from the colors it is indispensable to wet the book edges or the paper, to be marbled, with concentrated alum water, so that no spot is left free from moisture, because at that spot the color would not take.

Concentrated alum water is prepared in the following way; put 2 quarts of water and 13 ounces of alum into a pot and heat until the alum is entirely dissolved. This concentrated solution is permitted to cool and is bottled tightly.

A solution of alum in cold water is much too weak for the above purpose as alum is but sparingly soluble in cold water.

The book edge or paper, which is to be marbled, is wet with a sponge dipped into this cold concentrated alum solution and must be allowed to dry for from 5 to 8 minutes, after which the produced edge may be lifted from the size. Generally this point finds but too little consideration in our book-binding establishments.

Edges which have not been treated with this alum water show as a consequence poor and blotted results, although such little trouble is necessary to wet the edges with alum water.

With hair-veined edges, where colors are thrown on the size only in very small quantities, this wetting may be omitted; but with comb, peacock and bouquet edges, for which four to six colors are used, it is unavoidably necessary to prepare the edges with alum water because the glycocholate aluminium formed by the color while in connection with the alum water is essential to fix the color in such quantities.

Therefore, if it is desired to produce clean and lustrous edges in which the white lines appear without a trace of color, then this advice is to be followed, the small trouble being amply repaid by the beauty of the edges.

Ordinary earth and several of the lake-colors take without wetting of edges with alum water, but these colors leave much to be desired in their divisibility and fineness and always appear rugged.

A much finer fixing medium is the acetate of aluminium which can be easily produced.

Dissolve 1 pound of alum in 3 quarts of warm water, prepare another solution with the same quantity of soda, mix these two, whereupon the hydrate of aluminium is precipitated. Allow the latter to settle, pour the water above carefully off, and filter; the white body which remains consists of hydrate of aluminium which is readily dissolved by a little acetic acid, into a clear fluid and can be diluted with the same volume of water. Then heat the whole liquid, when the excess of acetic acid is evaporated, we have the desired acetate of aluminium which can be used in the same way as alum water.

Edges and paper which are to be marbled should only be moistened with alum water or aluminium acetate shortly before the marbling and should stand not longer than a half-hour after the sizing of the edges, because later on, an insoluble layer is formed which only takes color with great difficulty. It is therefore best to prepare the colors before commencing the sizing of the edges. If there are a large number of books, it is better to divide them into two or three lots.

The Preparation of the Colors for Marbling.



THE PREPARATION OF THE COLORS FOR MARBLING.

The want of knowledge of colors gave me the greatest trouble when I began their manufacture as I had no information as to why this or that color was unfit for marbling.

The numberless experiments I made to this end remained without results. It was an exceedingly great trial to my patience, but by the pertinacity with which I clung to my investigations, I, at last, succeeded in making the important discovery, that the real value of color for marbling purposes is its body and that the color itself, which is bound to the body either naturally or chemically, is of less importance. By this discovery every difficulty that I had so long experienced was cleared away and it was made manifest, why it had been impossible for the marbling art to become sufficiently popular.

From this time on I had a foundation, upon which to continue my investigations with greater surety. Finally I succeeded in finding that the colors named later on are good and useful marbling colors.

When we look over the different special branches of our trade, we find everywhere great progress, which is even noticeable in the smallest workshop. Already, these wonderful achievements have become common property, only the art of marbling stood still on account of insufficient instruction and it has not made any progress in the last decade. This was due to a large number of men in our trade using mechanical marblers to avoid those obstacles, which had hitherto offered themselves in marbling.

The expectations which were placed on the mechanical marblers were not fulfilled, because, after a short use, and when not carefully kept clean, they became sticky from dust and color mixing so that they gave but very indifferent productions. The necessity of continually cleaning these mechanical marblers and their lack of uniformity are by no means commendable properties.

Mechanical marblers should be retired to small book-binderies for use upon single books, where they are eminently in the right place, driving away the primitive sprinkled or starched edges.

The interest which was shown in mechanical marblers is a proof that all members of our trade, who use them, possess a liking for the marbling art and would be zealous friends of the same, if they but knew of the methods of marbling, that would give beautiful results without great difficulty.

To judge rightly the value of a new invention, we have only to see, whether it is introduced into the wholesale trade and is in continual use. This is the best test of inventions and the only proper basis, upon which to judge them correctly. We find the mechanical marblers exclusively in the small shop, because they do not answer the demands of the wholesale manufacturer in any way.

What the marbling art is able to offer, a person can only judge, who has seen exemplary edges and to whom marbling is not a stranger. No kind of edge, to which color is applied, can be compared to the effective splendor of marbled edges. In thousands of variations it imitates nature and delights the eye by its products. Although marbling has hardly reached the middle rung of the ladder of its development, it to-day stands unrivalled.

It is clear, that such an art should be fostered and nursed, if it is to grow to its full development. But here the investigation of one man is insufficient, a general interest is demanded and I hope, that this will be aroused by my work. After this slight diversion I will now begin to treat my real subject, the preparation of colors, and for that purpose, I will divide the marbling art into three important parts: the size, the color and the technical application.

The size and the technical application of the colors demand great attention and practice; on the other hand, the preparation of colors but labor. The colors are divided into two groups, mineral and lake colors. In the lake colors, if they are good and strong, the coloring matter bound to the metallic oxides is equal to one-half of the whole weight of the color, therefore they can be prepared as good marbling colors without adding anything but gall. On the other hand, in mineral colors, which in a natural or chemical way are bound to bodies, the body weighs much more than the coloring substance and even if the metallic oxide is the real coloring substance, as is the case

with ferric hydrate, the organic coloring matters derived from the vegetable or animal kingdom are preferable to the inorganic substances.

Why this is the case we shall see further on. In those coloring substances, which I have recommended as good marbling colors, the aggregate state of the body is amorphous and admits of the greatest divisibility. If such a coloring substance is ground in the finest way possible, the division of the body is a very great one, but an intimate union of the particles can never take place because first, the color is insoluble in water and, second, because the division (diminution) of a body has its limits. For this reason particles are isolated and lay loosely alongside of each other.

If a mineral color is mixed with the necessary amount of water and gall and thrown upon the size the drop expands to a round disc, which upon close investigation, is found to consist of numberless small dots. When such coloring stuffs are used, the union of the particles must be brought about by some glutinous matter. This is unnecessary with lake colors, where the coloring matter is as heavy as the body to which it is bound, as all organic substances possess a sufficient quantity of glutinous matter to form a union of the particles.

The clearest proof of the correctness of my allegations we find in a mineral color, in which the metallic oxide was precipitated by a semi organic salt, I mean French blue, which without any substance unites the particles quite well if the washing of the precipitate after the blueing, was a perfect one.

The glutinous matter used with mineral colors must be entirely free of any acids as the smallest particle of acid, which in warm weather is often formed after long standing causes an isolation of the coloring matter in such a way, that it will become curdled when thrown upon the size. The best glutinous matter, that can be used, is dissolved isinglass or parchment-glue with gum tragacanth. As the first two named materials are expensive they can be replaced by fine gum arabic free of acid. Take $\frac{4}{6}$ of an ounce of gum tragacanth let it swell in 1 pint of water and dissolve fully by boiling. In this way we obtain a glutinous mass, which is mixed by stirring with a thick solution of gum-arabic. This mixture is used to unite the particles of the body of mineral colors.

To grind the colors use a very smooth marble slab, a roller and a palette knife, of wood or horn, as those of steel produce some change upon red colors.

The colors which are sold in the market are either in the form of grooved pieces, little cones, or powder. If the color is in pieces or little cones, it is first ground to powder in a dry state and is then mixed with a little water and from 10 to 15 drops of ox-gall to a thick paste. If a mineral color, add a piece of the glutinous matter the size of a nut, if, on the contrary, we have a lake color, the addition of water and gall is fully sufficient.

Now commences the grinding. With a muller you continually run in a circular motion over the color using a moderate pressure of the hands. After about two or three hours have elapsed the color will have the requisite fineness which is shown by its greasy, lacquer-lustrous appearance. During the grinding, move the color from time to time into the center of the marble with the palette knife and at the same time the adhering color is removed from the roller.

When the color is ground fine it must be diluted with water in a glass vessel and is then ready for use. Bottle tightly and keep it in a cool place, best in the cellar.

For grinding the colors and diluting the same, river or rain water should be used, of course, proportionately to the quantity of color so that the latter will not become too thin.

The grinding of a color is done mostly by apprentices as the time of a journeyman would be too valuable. In most cases this part of the preparation gives the greatest trouble, since the colors are seldom gotten to that fineness necessary to marbling. This is easily explained as the grinding demands practice and a certain perseverance. From the point of view that time is money, the majority of book-binders concluded to use ready ground colors and this was profitable to them as they could buy them better and finer ground and much cheaper. But, even here, we sometimes find articles, which leave much to be desired, but which, in most cases, can be used.

For fifteen years I have been carrying on the manufacture of marbling colors and must always endeavor to keep pace with the progress of the

chemistry of the colors as new products are continually offered for sale superior to the old in quality and beauty.

To give to the trade an idea of the manufacture of marbling colors I will briefly describe it.

It is a well known fact that among the great number of colors produced by a factory, there are always several, which excel by their especially good quality. I selected, therefore, from among the colors of several factories, those which were especially excellent and I now control an assortment, which hardly any single factory is able to supply.

In preparing them 20 pounds are always ground at once. The color is ground upon a large marble slab with water and gall, to mineral colors the necessary glutinous matter is added and this is formed to a thick paste and then ground in the color-mill. The grinding in the mill is essential to obtain a homogeneous mass and then this paste is ground twice on a color grinding machine with three porphyry rollers in the finest possible way.

The color having been diluted is then prepared on a size, which is gotten up for this purpose and then preserved in stone jugs.

It is easily explainable, that the best results must be obtained with such a carefully and excellently prepared color, I, therefore recommend to my fellow marblers, the marbling colors manufactured by me. [\[A\]](#)

Before using, the color must always be well shaken and there must only be taken from the bottle enough necessary to one marbling process. The remaining portion of the color must not be poured back into the bottle, because if this be repeated several times the whole color would be spoiled by the gall.

A small quantity of, say from 40 to 50 drops of color, is sufficient for marbling the edges of even a large number of books, the colors must always be kept tightly bottled up as they may be spoiled by being exposed to the air for a longer period.

The Marbling of Book Edges and Paper.



THE MARBLING OF BOOK EDGES AND PAPER.

The Marbling consists, generally speaking in this, that finely ground fluid colors are thrown in drops by the aid of brushes on a thickly fluid size contained in a flat trough, that then by the aid of gall, different marble-like designs are formed, that they are lifted off the size by immersing the edge of the book and by that are transferred on this edge.

Before speaking of the different kinds of edges, I will mention something of the harmony of colors, as in drawn edges the real harmonious combination of colors is of great influence on their beauty. The application of color in the art of marbling gives us certainly extraordinary freedom and a wide range to our fancy, but nevertheless there are certain principles or laws necessary to make the colors appear in an effective and beautiful way. Those tones of color which lie between yellow and reddish yellow are called warm, those between blue and the middle of reddish blue, cold colors. In the center lies red by its strength and effectiveness forming the dividing line between the warm and cold.

To explain the principles of color harmony, the colors are divided into three groups, first, primary colors, second, secondary, and third, tertiary.

Primary colors are those which cannot obtain their inherent purity by any mixture, namely red, yellow, blue, white and black. Secondary are those which are generated by a combination of two primary colors, for instance, orange, purple, green and grey. Tertiary, or broken colors, are those which are obtained by mixing two equally powerful secondary colors, for instance, brownish red, olive, slate grey, etc.

In the harmony of colors luminosity and intensity play the main parts. The darkest color must occupy three times as much space as the lightest and the succession must be kept up in this way always taking into consideration the luminosity of the colors.

Which colors must now be placed alongside one another to produce a harmonious effect? The laws for this are founded on the peculiarity of the power of man's vision. The eye, demands to be satisfied, an accord of primary colors. Red requires to annihilate one sided irritation the two primary colors, yellow and blue. The secondary green is obtained by a combination of yellow and blue, for that reason green is the color demanded by red to please the eye. But there must be a contrast of colors, for instance light red and dark green or dark red and light green must be brought together. The same thing is the case with other primary colors. The primary color yellow demands the secondary purple, blue, the secondary orange, the tertiary color olive, the secondary orange, brownish red and green, brimstone yellow and violet.

The correct treatment of the shadow tone of a given color depends upon the knowledge of the laws of the annihilation of color effect. As in nature, a shadow is formed by obstructing the light, so in the technic of coloring the shadow is obtained by the annihilation of the light-color. By combination of a primary color with a harmonizing secondary color of equal strength, we obtain the right shadow tone of the first color. In this way, by experiments in mixing, the right shadow tones, which form in a multi-colored edge, the gradual transitions of primary colors can be easily detected.

THE COMB MARBLE.



THE COMB MARBLE.

(SEE [SPECIMEN PLATES.](#))

The comb or nonpareil marble belongs to those edges which are not so frequently used, although it is superior in brightness and variety of colors to other marbles, provided the combination of colors is a correct one and there are no entire blotches of the same color present. For the comb-marble in general use, four colors are mostly employed, namely black, blue, yellow and red. To these white is added, but not as a color, as it is formed by the stylus in drawing the colors.

For precaution's sake let me here repeat the description of the preparation of colors: before using them shake well and then pour into the different small vessels as much of the same as is necessary for one marbling operation only, then put them away handy for use in the following order, black, blue, yellow and red. Black is used for shading all other colors, therefore, it should be thrown on first as the ground color. As I mentioned before, the colors are to be prepared upon a small surface of size so that the whole size may not be soiled, as it is nearly unavoidable during the preparation to prevent several drops from sinking. Whether the colors are suitable to each other as regards their expanding power or whether the size is too thick or too thin for comb-marble is determined in the following manner; draw off the size in a dish, by a strip of paper, throw on a drop of black, which if size and color are correctly proportioned must expand to a spot of 4 inches in diameter. If the drop does not expand in this way the size is either too thick or more gall must be added to the color. Add from 5 to 10 drops of gall to the black color and repeat the experiment. If then the color does not expand to the given diameter take a stylus and draw it in wavy lines such as used for comb-marble. If the color can be drawn into beautiful straight lines, without following the stylus, the size possesses the normal consistency and more gall must be added to the color, until it reaches the aforementioned diameter, but if the color is drawn along by the stylus or cannot be easily cut by it, then the size is too thick and must be carefully diluted with water. If, on the other hand, the color in drawing with the stylus should run, then

the size is too thin. This can also be easily seen by the extraordinary quivering of the size. Such a size cannot be used for comb-marble, either you must add some thick size or you must use it for common marble edges. The black color is the key to the success of a beautiful edge; the other colors must be prepared according to the strength of the black.

When the ground color has spread out to the desired dimension, throw on the black a drop of blue, then yellow on the blue and red on the yellow which must all expand to a diameter of 1-1/2 inches. Should this not be the case then several drops of gall are to be added in the order aforesaid. The three latter colors must be prepared by themselves, and if, after throwing a drop of blue on the black, the former does not reach the necessary expansion, gall must be added in drops until this is achieved, then the yellow is manipulated in the same way.

When all colors are toned in regard to their power of expansion, then and not before then, can the marbling commence.

With a little experience or practice the whole manipulation is finished in less time than it takes to describe it.

Pour the size into the trough, take the top of it off with a thin piece of wood which fits into the trough closely, smooth it over with the wood so that the surface is free of bubbles, then throw on, by the aid of a hair brush, the black in the direction of the length of the trough towards the centre of the size in such a way that the first drop comes in contact with the rim of the second, the second with that of the third, the third with the fourth, etc., so that a ribbon from 4 to 5 inches wide is formed. Then throw on the blue in drops at both sides of the black following the length of the trough, but the joining of the drops must be avoided. Yellow is thrown on at both rims of the black just as the blue was but in such a way that each drop of blue has a yellow centre. Finally red is thrown on the yellow so that each drop of yellow receives a red centre. By this means we produce a beautiful combination of colors. Then take a thick knitting needle or a thin stylus of wood and draw the colors in wavy lines through each other but so that the stylus will pass the boundary of the black. By doing this white lines will be produced between the colors, adding at the same time, the fifth color white

which causes the marble to attain a brighter aspect. If then the comb is drawn across, beautiful scales will be produced.

The books which have been wetted shortly before with alum water, are now placed between boards or clamps and are dipped from right to left, somewhat obliquely, into the trough so that the left end of the book touches the size first, and then with a firm hand the book is inserted towards the right until the right end has touched the size. This must be quickly done to avoid the formation of air bubbles. As soon as the marble edge is taken off, the superfluous size is carefully removed from the edge by a moist, soft sponge so as to prohibit moisture from penetrating too much into the leaves of the book. In this operation great care must be exercised to prevent the marble edge being rubbed out or spoiled.

The remainder of the color is to be taken off the size by the aid of strips of paper and this is done in the following way: one strip is inserted into the size at the left end of the trough in such a way that it slightly enters the size, then by the aid of a second strip of paper, the top of the color is taken off beginning at the right end of the trough and moving the film to the left till it is brought between the two strips of paper and can be removed from the trough into a basin which is kept ready, to receive the waste color. If the colors are quickly removed in this way, the size itself remains perfectly clean even though a large number of books be treated.

On good work, the book at first is only cut in front. As soon as the front edge is marbled and dry the book is backed, and cut at the top and bottom; both edges are wetted with alum water and again the colors are thrown upon the size taking care, that the scales of the upper edge run in the same direction towards the fore-edge as those of the lower one. In more common bindings, all three sides are cut together and after the front edge is marbled and the book is rounded and backed, the marbling of the upper and lower edges is executed without delay.

The rounding of the book must be done before the marbling of the upper and lower edges, because the comb would become disarranged, if the rounding were executed afterwards. Remember that always before throwing on the colors, the top of the size must be taken off by the aid of the piece of board, as, by evaporation, a film is formed over the surface which does not

permit the expanding of the colors. The quicker the marbling is executed, so much more beautiful will be the edges.



PEACOCK MARBLE.



PEACOCK MARBLE.

(SEE [SPECIMEN PLATES.](#))

Peacock marble is one of the newest and prettiest of marbles. It has been much used in recent times for end papers of books, which receive the same edges and in this connection gives a beautiful effect. The method of preparation is generally the same as that of the comb-edge, the difference being this, that, after the colors have been drawn by the stylus into wavy lines, it is drawn by the aid of a movable comb, (the preparation of which I shall describe among the tools) in such a way, that for a distance of 1/2 inch the comb is made wider and again after a distance of 1/2 inch is made narrower. This is continued until the entire marble is crossed by this opening and closing comb; of which a trial is all that is necessary to make it all clear.

THE BOUQUET MARBLE.



THE BOUQUET MARBLE.

(SEE [SPECIMEN PLATES.](#))

There is no doubt that from among the drawn marbles the bouquet marble is one of the most brilliant, as it can be compared to a number of buttonieres placed alongside each other, if the combination of colors is a good one.

In the manufacture of colored paper, the bouquet marble is mostly produced from two or three browns and one black, and is often used for lining end papers in fine books.

After the colors have been thrown on, drawn by the stylus and combed in the ordinary way, a rake-like wooden instrument is moved through the colors right and left across the whole surface of the size so that in moving the rake the teeth of the second row trace those of the first row precisely, and so on until the rake has been moved over the whole surface from the right end of the trough to the left. A somewhat changed design is produced by treatment of the colors in the same way leaving out the marbling comb and spreading the drops of color thrown on only by the aid of the stylus into very narrow cross-lines and then using the aforementioned rake as already explained. The rake consists of a piece of board of hard wood, of about 1 inch less in length, than the width of the trough, 1-1/4 inches wide and 1/2 inch thick, into which are inserted two rows of sharply pointed wooden teeth about 1 inch distant from each other and leaving a space of about 1 inch between the two rows, in such a way that the teeth of the second row are situated precisely between the teeth of the first row.

ENTWINED COMB MARBLE.



THE ENTWINED COMB MARBLE.

(SEE [SPECIMEN PLATES.](#))

This comb-marble, produced by drawing the comb back, is a great favorite on account of its beautiful designs, but is more adapted for thicker books as on those only can the pattern be seen to full advantage.

The production is entirely the same as that of comb marbles, but the teeth of the comb must be about 1/2 inch distant from each other and the comb must be 1/2 inch less in width than the trough so that it can be moved right and left at will, when the marble is drawn by this comb the latter is drawn back in such a way that the teeth cross the centre of the scales.

THE SNAIL MARBLE.



THE SNAIL MARBLE.

(SEE [SPECIMEN PLATES.](#))

The production of this marble is very simple. The colors are thrown on as for comb-marble and are drawn by the stylus into wavy lines and then, one by one, by the aid of the stylus, snail forms are produced.

THE GRAY SNAIL MARBLE.

The manner of production is the same as that of the plain gray marble, with the exception that so much green is added till a desired tone is obtained. After this ground-color has been thrown on, the marble is drawn by the aid of the stylus into snails and then taken off.

THE COMMON GREENISH-GRAY SNAIL MARBLE.

To produce this the colors are thrown on in the following order, black, light-blue (two parts water, one part Oriental blue,) Indian yellow, pink, light red, gall water, and, finally, the body color, composed of one part black and two to three parts sprinkling water according as the demand is for a lighter or darker body color. Throw the colors on the same as in producing the grey marble, (see [marbled edges](#)) and when the ground color is thrown on draw the marble into snails and take off.

THE DARK BLUE SNAIL MARBLE.

To all the dark marbles, which should have an especially brilliant effect, and are adapted for light-colored bindings, there is always added one or two parts of a solution of shellac and ammonia to the body color, in addition to sprinkling water. To produce this marble take black, light brown, light blue, (one part Indigo, two parts white,) lemon yellow, gall water and the body-color, which consists of two parts Indigo, one part black, from one to two

parts sprinkling water and the same quantity of shellac-ammonia solution. After everything is thrown on the snails are formed.

THE DARK-RED SNAIL-MARBLE.

This marble has a beautiful effect and is especially adapted for fine half bindings with light-colored leather backs and corners and end papers of the same pattern as the edges. To produce this the colors are thrown on in the following order, black, light red, pink, light gray, (four parts white, one part black,) gall water and the body color consisting of two parts of carmine lake, one part of black and two parts of sprinkling water and shellac-ammonia solution. After throwing on the ground color the snails are drawn.

THE GRAYISH-GREEN SNAIL-MARBLE.

To produce it the colors are thrown on in the following order, black, claret red (two parts carmine lake and one part of black well mixed,) pink (ten parts of white and one part of scarlet red) gall-water which forms the white veins, and, finally, the body color, which consists of one part of black, three parts of sprinkling-water and an addition of green according to what shade of green is desired.

GENERAL REMARKS UPON DRAWN MARBLES.

GENERAL REMARKS UPON DRAWN MARBLES.

All marbles that are drawn by the stylus or knitting-needle are, in their treatment, more or less similar to the comb-marble, therefore the way to produce them is similar to that described in the former article upon comb-marbles.

To produce a pretty coloring, it is unavoidable, to use those four colors which I have named in the former article on comb-marble.

If it is desired, to use more colors than the four named, greater perfection in marbling is essential and it takes a perfect practical marbler to achieve beautiful products with eight colors, but practice and a zealous heart will even overcome these difficulties. The lustre of the colors is due to their combination, according to their power of spreading out and to their harmony.

Wall-paper will frequently show what beautiful and fine effects can be produced by three, or at the most four colors.

If the colors are separated by white, they develop a greater lustre, while black employed in the same way is productive of a more sombre effect.

The separation of colors by white and black is most decidedly more profitable and effective, than a marble in which the different variegated colors immediately touch each other. It is therefore advisable always to take black as the ground-color, white will be produced without adding any coloring matter whatever by the drawing of the colors with the stylus.

Black is the ground-color, upon which all the rest, blue, yellow, red are thrown; it is understood, that they must expand in the proper proportion, to produce a clear marble.

The lustre of the colors is mainly a result of white and black, which form, as it were, a frame about the other colors by their own durability.

The other colors, which we may add to the four nonpareil colors, are, different shades of red and yellow. A bottle is filled to one half its height with carmine-lake, then black is added until a very deep purple is obtained; another color is a powerful orange, which, if not on hand, can be produced by mixing red and yellow. To produce a beautiful combination use the following colors, black, light blue, orange, light yellow, purple and finally scarlet red, (a color which I have recently introduced under the name of safflower carmine) this combination of colors gives a surprisingly beautiful result. To produce the nonpareil (or comb) marble with eight colors, two more mixed colors are used, one of which serves as a shade to light blue, while the other enhances the lustre of the lighter colors. To this end, pour a little dark-blue into a glass bottle and mix it with the same quantity of green, obtaining a dark bluish-green. As a second color mix green and white until the color is pale green. To produce a desirable effect with these eight colors we must observe the following order, black, light blue, dark bluish-green, orange, light yellow, purple, pale green and scarlet-red.

In this wise numberless variations and combinations can be obtained, but it should always be taken care, that the primary colors only receive shading tones of secondary or tertiary colors. The mode of throwing the colors on the size is always the same as I described it for nonpareil marbles namely; form a ribbon of black from 4 to 5 inches in width in the prescribed way and throw the other colors into the black and at both rims of it but so that they are situated within the black. The same colors which are on one rim must also be thrown upon the other rim in the most uniform way possible. The marble therefore, before it is drawn is similar to a ribbon which has in its centre the black stripe about 2 inches wide, and on each side, a border of different colors 1 to 1-1/4 inches wide.

The more colors are used for the drawn marble, the less of each color except black should be thrown on the size. Although the colors have been adapted to each other in regard to their power of expansion before using them, it is indispensable in producing these marbles with such a large number of colors to assist the expansion of one color or the other by a few drops of ox-gall.

It is necessary to see, that the second color should not too greatly be displaced by the third, the third by the fourth, etc., because the color, which

is mostly displaced would not appear in the marble at all.

In producing peacock marbles, I obtained most excellent results even without primary colors. They were not so bright and lustrous in their general color effect, but their most subdued tones were most pleasing to the eye.

To one of these marbles I used the colors in the following order; black, medium olive green, dark bluish-green, bronze or ochre yellow, dark blue and bright chrome yellow, (to obtain the olive, use the following mixture; 3 parts light orange, 1/2 part dark blue; bluish-green as described above; ochre yellow, 2 parts light chrome yellow, 1 part brown; blackish-blue, 2 parts dark-blue, 1 part black; light chrome yellow, 2 parts light orange, 2 parts white.)

Beautiful bouquet marbles are obtained from the following colors; blackish-green, 2 parts black, 1 part green; yellowish-green, 2 parts yellow, 1 part green; light-brown, dark bluish-green as above, scarlet-red and light ochre yellow. In this way many combinations of color can be obtained, and it is left to the judgment of the marbler to select colors and make the mixtures. Having succeeded in producing beautiful marbles with few colors, it will be easy for anyone to produce others with more colors.

MARBLED EDGES.



MARbled EDGES.

(SEE [SPECIMEN PLATES.](#))

Many of my colleagues are of the opinion, that marbled edges are inferior to drawn edges, but this is a great mistake, and here we are able to show in what directions the marbling art is able to develop to its full glory.

The field for marbled edges is so large and prolific in the variations of shade, that there are really no limits to its capabilities.

While making experiments in this field, by accident I came across some edges without having an idea of the effect of the combination which so astonished me by its beauty.

It would be endeavoring too greatly to describe the method of producing all effective marbled-edges which came to my knowledge in my many years' experience, I therefore name but a few which will certainly invite imitation and study.

Marble is the design of an edge of a book, which has a ground or back color and over which a net of veins of different colors is stretched.

For the production of marbled edges, the following preparation of size is advisable and in general use: boil 7 quarts of water and 3-1/5 ounces of carrageen moss, after boiling add 1 quart of cold water, in which one ounce of common soda has been dissolved, allow it to stand for 12 hours, and filter it through a linen cloth when it will be ready for use.

The preparation of colors is effected as with drawn marbles on a small surface of size, but it is necessary, that the colors for marbled edges expand more than the others. As first color, black is generally used, which is thrown on by a brush in the same way as in making nonpareil marble, but the other colors must be thrown on by a broom-corn whisk and the more colors are used, the smaller must be the drops of the colors which are thrown on the size, and it is further essential that they should be evenly divided around. Then use gall-water which consists of 10 parts water and 1 part gall and

serves to arrange the different colors so that one color can be concentrated in one point, secondly, it has the effect of forming a white network of veins, which enhances the effect of the other colors, and finally, the ground color comes into play, putting the whole carpet of colors into motion. This color is nothing but a common nonpareil color and is used either by itself or in a mixture with other colors.

In preparing the ground or body color, enough of gall is added so as to slightly force the other colors into veins. An equal part of sprinkling-water is added to it, to give it the energetical power of spreading out essential to the formation of a beautifully veined net. The body-color is thrown on by a medium sized bristle brush, which must be tied near the end for this purpose. This brush is to be equally moistened with the body color, the best would be to moisten it by the aid of a common brush. When this is done throw the color by beating the bristle brush on a small board upon the carpet of colors. By this process larger or smaller specks of the body-color are formed.

THE DARK-RED MARBLE.

The marbled edges are always named from the body-color, as the latter always occupies the largest space on the edge. For dark-red marble the following colors are used: black, light pink (10 parts white, 1 part safflower-carmine), light-grey (10 parts white, 1 part black), gall-water and, finally, the ground color.

The latter is composed of 2 parts carmine-lake, 1 part black and 3 parts sprinkling-water.

THE PLAIN-GRAY MARBLE.

This kind of marbling is the most profitable of the veined marbles because it can be produced very rapidly.

For this purpose prepare the black color with gall in such a way, that a drop will spread out on the size to a spot of about 5-1/2 inches in diameter, then add an equal quantity of sprinkling water, dip a large broom-corn whisk or a bristle-brush into the color and throw it by beating on a board upon the size

which has been cleaned off by a strip of paper. The color appears on the size a gray color with white veins, from which the edge can be taken.

THE BLUE MARBLE.

To produce this use the following colors; black, orange or citron yellow, Indian red with a little black and Oriental blue, finally adding gall water and the body color. This ground color consists of 1 part Indigo and 1 part sprinkling water.

THE BROWN MARBLE.

This marble is one of the most common, but if the colors are well selected as given in the following paragraph it will give a fine effect.

Black first, Havana brown second, chamois third, gall water, and, as ground color, humin-brown with a little carmine lake and one part sprinkling water.

THE GRAY MARBLE.

To produce this the following colors are necessary, black, dark purple, light bluish green, gall-water and a mixture of black with sprinkling water, by which the ground shade, gray, is obtained. A lighter or darker gray is obtained by adding more or less sprinkling water to the ground color.

THE OLIVE-GRAY MARBLE.

This marble has a very beautiful effect although in producing it but three colors are used. It belongs to the so called large-veined marbles and forms an olive grayish ground with a fine net of veins of black, dark-brown and white. It is produced in the following way; first black is put on with the brush, then brown is squirted on by the aid of a broom-corn whisk and then gall water and finally the body color is added. The body color consists of 2 parts black, 1 part green and the necessary sprinkling-water. If the shade of the body color is too deep, some pure water is added. For producing large

veined marbles, the brush must be well filled with color, so that in throwing on of the color quite large drops will fall.

THE BLACK MARBLE.

It was formerly an unknown thing to produce a black marble over which a net of light-colored veins was stretched. My investigations have resulted in my becoming able to produce this brilliant and effective marble.

The colors used are scarlet-red, bluish-green and gall water, the latter representing the white, this network of veins comes out excellently from the dark ground-tone. The scarlet-red color is first taken and must spread out in a like way, as the black in the other two marbles. Then follows bluish-green and gall-water and finally the body color is added. The latter makes a special preparation necessary. Take $\frac{3}{5}$ of an ounce of ivory-black add $\frac{1}{6}$ of an ounce of a thick solution of gum, grind the whole mixture with a mixture of shellac-ammonia which has been diluted with a little water. If the color is ground perfectly fine it is diluted with the solution of shellac-ammonia and is ready for use. Should it not be powerful enough to force the colors into veins, a few more drops of gall must be added. This body color is likewise thrown on by the aid of the large bristle brush as all other body colors.

As beautiful as this kind of marbling may be, there is a disadvantage in its use as the body color must always be newly prepared, because on the second day after standing it curdles, and cannot be used any more.

THE BLUISH-GRAY MARBLE.

By mixing the colors very beautiful effects may be obtained. One of these fine marbles is the bluish-gray. To prepare it the following mode is used; claret red (2 parts carmine-lake and 1 part black) and as body color, Oriental blue mixed with black and sprinkling-water.

GRAY MARBLE WITH A NET OF COLORED VEINS.

To produce this the following colors are used; black, yellow, cinnabar-red or Indian-red, Oriental blue and gall water and finally the gray body color.

The colors are put on in the following way; first black with the brush into a ribbon-like form, then yellow is squirted on in small drops by means of a broom-corn whisk, then follow red, blue and gall water which are treated like yellow, and finally, the body color is squirted on by the aid of a whisk or the large brush. The latter forces the colors by its great spreading power into beautiful veins, after which the edge can be taken off. The body color consists of 1 part black and 2 parts sprinkling water.

THE GRAYISH GREEN MARBLE.

This is one of the most beautiful specimens of marble and according to the shading of the body color most beautiful effects can be produced by it. The following colors are used to produce it; black, carmine-red (1 part carmine lake and 1/2 part black), gray (1 part black and 10 parts white), finally gall-water and the body color.

The mode of producing it is the same as described before. The body color consists of 1 part black, 2 parts of sprinkling water and as much green as is necessary to produce the desired shade.

Before adding sprinkling water to the body color, black, only so much gall is added that it will but slightly push the different colors thrown on.

HAIR MARBLE.



THE HAIR MARBLE.

(SEE [SPECIMEN PLATES.](#))

These marbles are in great use and are much liked on account of their rapid and easy production, especially when many books are to be marbled; they appear very elegant on account of their fineness.

For hair marbles but one or two colors are employed which are diluted with water to $\frac{1}{3}$ their volume. Add enough of gall to the first color so that in throwing the same on the size it will spread sufficiently to form a ribbon from 5 to 6 inches wide. Take a whisk and throw on the second color in drops that are not too large and as nearly as possible, equally distributed, then, with the bristle-brush, or sieve and brush throw sprinkling water on the colors. In this way hair veins are formed, showing more or less ground according to the quantity of sprinkling water thrown on.

The size is produced for all hair-marbles alike. In the morning boil 3- $\frac{1}{5}$ ounces of carrageen moss in 7 quarts of water and after the boiling add 1 quart of cold water, stir the whole mixture several times and allow it to stand during the day until it is fully cooled off. In the evening the size must be again diluted by another quart of water and it is then filtered through a piece of linen and left to stand over night. The next day it is ready for use. This holds good during the summer months as the temperature of the air will just make the size useful for this purpose. During the winter months the size must be allowed to stand for 36 hours before use.

BLACK HAIR MARBLE.

To produce this variety of marble there is a screen or sieve necessary as well as a wide brush having stiff bristles. By adding gall from time to time, the black is made so expansive that a single drop will be spread out to a spot of about 8 inches in diameter. Dip the brush into sprinkling water, shake the brush well so that the superfluous water is thrown out, hold the

sieve above the size and move the brush quickly and lightly over it. In this way the hair marble is finished and may be taken off.

RED HAIR MARBLE.

For this to be produced the best color is Indian or mineral red since all other chemical coloring substances which are bound to bodies will usually run, on strong compression, into hair veins. Only the excellent lakes of Munich will withstand this treatment, but the fineness of the veins leaves much to be desired, whereas Indian red answers every purpose.

BLUE HAIR MARBLE.

The mode of producing it is the same as applied in producing black hair-marble, but, instead of black, Indigo is used. Indigo is the only one among blue coloring matters that admits of the necessary expansion for the production of hair veins without running off or gathering into small lumps.

RED AND BLACK HAIR MARBLE.

For the production of this marble, black and Indian red are taken. The black is first thrown upon the size as in black hair-vein marble, the red is then treated by adding gall until a drop thrown upon the black will spread to about 2 inches in diameter. The trial is made with a brush, but for the purpose of producing the marble a whisk is taken, by means of which the red is thrown upon the black in very small drops and close to each other.

BLUISH-BROWN HAIR MARBLE.

To produce this marble Indigo and Vandyke brown are taken. The Indigo is put on first then the Vandyke brown is thrown on by a whisk, and finally, the carpet of colors is forced into forming veins by sprinkling water, which is applied by aid of the brush and sieve.

Rules in regard to the Marbling Table.



RULES IN REGARD TO THE MARBLING TABLE.

The table upon which marbling is to be executed must be firm and immovable. On it the utensils necessary to marbling are placed. The trough, colors, gall in a closed bottle with stopper pierced by a tube, a vessel with water, the comb, the size skimmer are all to be within easy reach. They should be placed about in the following order; in front, nearest to the workman and to the right, the trough, the receptacle for the waste color and board for skimming off the size to the left, beyond the trough the vessels in which the colors and gall are kept, behind the first vessels containing the colors the dish containing water, and alongside this the bottle containing gall and next to it the comb and stylus. On the left side of the trough there must be a sufficiently large space for the books to be marbled.

PRECAUTIONS.

Before closing the article on marbling, I will add a few words to the precautions to be observed while marbling. Before all else, see that the air in the room, where the marbling is to be done, is free of dust, as the falling dust, especially in drawn marbles will form many white spots, which spoil the effect of the marbling.

Cleanliness should always be observed, as the beauty of the marbles and their rapid production depend on it. Protect the size from dust by a lid, which you place on the vessel, and have a large soft sponge ready to remove the superfluous size after each immersion of the book, by which a rapid drying is obtained and the moisture is prohibited from impregnating the book.

SUPPLEMENTARY REMARKS.

A NEW EXPANDING MEDIUM.

I must mention here a newly discovered medium which has been found essential in the formation of beautiful veins in plain marble; take 1/10 quart of shellac orange, add 1/8 quart spirits of ammonia, allow the whole to boil for 5 minutes stirring repeatedly (by which a uniform solution of shellac in ammonia is obtained), take it from the fire and dilute the solution with about 2 quarts of water. Throw the fluid by means of the brush on the size and then throw on the body-color, see [plain gray marble](#). In this way beautiful white veins will appear, which is explained by the fact, that the surface of the size is covered with an expanding medium, which can be forced together by the body-color, but is strong enough to hinder the union of the latter.

MARBLING OF GILT EDGES.



MARBLING OF GILT EDGES.

If gilt, silver or aluminium edges are to be marbled, burnish slightly and moisten by a soft brush with a solution of powdered alum in hot water which is of the consistency of a thin pap.

The marbling of the gilt-edge is then executed in the usual way before the coating of alum is entirely dry.

After the marbling is done, water is poured over the edge until it is entirely clean.

When the edge is completely dried it is rubbed over with a waxed rag and is then burnished in the usual way. For marbling the usual marbling colors are applicable, but it should here be added, that very dry edges will not take the marbling colors readily.

MARBLING ON A LARGE SCALE.

For the purpose of marbling on a large scale especial arrangements have to be made, and it is wonderful how elegant and practical are the marbling rooms of large establishments. For instance the tables and floors are of stone and cement, or of wood covered with zinc plates, and the troughs and receptacles to take off the fluids which have been employed are to be found in universal use. Even large iron cranes are erected to hold the books together and lift them off, and to transport them to the revolving drying apparatus, which can be heated.

These arrangements are in keeping with the extent of the business and can here only be lightly passed upon.

THE UTENSILS.



THE UTENSILS.

THE MARBLING TROUGH.

The marbling trough is a long flat oblong box of about 20 inches in length, 10 inches wide, and 1-1/4 inches deep inside, and should be painted within with white oil-paint so that the size appears light making the colors more readily discernible.

This trough must have inserted on one side a sloping partition of sheet zinc so that the waste of color may be skimmed over it by aid of paper slips or a piece of wooden board without running it over the top.

In addition to the trough containing the marbling size, a second smaller trough or basin of 10 inches width, and 1-1/4 inches depth is necessary for the reception of the waste of color. This is to be placed, while marbling, immediately to the left of the first trough.

SIZE SKIMMER.

This is a piece of wooden board about 2 inches high, about 1/4 inch thick and of the exact width of the trough. At both ends the board is slightly wider than the trough, protruding, and while the larger part of the board moves over the surface of the size, the projecting ends act as guides by sliding over the rim of the trough. In this way the board is always kept in a like position and serves to remove the air bubbles and films and also the color waste.

Recently, strips of blotting paper 1 inch wide have been frequently and successfully used for the removal of waste of color. They are applied in the same manner as the board.

THE STYLUS

of wood or bone, is used for the drawing of the colors in producing nonpareil marbles. This was described in the chapter on nonpareil edges.

THE COMB.

For the production of nonpareil marble, the comb is employed, made in the following way; take two strips of heavy pasteboard, 12 inches long and 2-1/2 inches wide, divide one of these exactly where the needles are to be inserted, draw grooves so that the needles can be laid in; glue the part in which the needles are to be laid, insert the needles in the grooves as aforesaid, and, after drying, the second strip of pasteboard, is glued upon the first.

The needles best adapted to comb making for use in comb marbling are fine pearl needles of steel; they are preferable to the common pins because they do not bend and do not offer any resistance to the size while drawing the colors, and also because they cut the colors easily without forming furrows.

The comb is so prepared that the pasteboard protrudes slightly at both ends, by this the movement of the comb is defined by the rim of the trough. It is advisable to prepare about three such combs with the needles separated from each other about 3/32, 3/16 and 1/4 inch respectively.

THE PEACOCK COMB.

The peacock comb consists of two common combs having strong needles separated from one another by 1/2 inch. These two combs are put together by means of a pasteboard frame in such a way that they can be moved 1/4 inch to the side.

Moved sideways the needles of one comb move to the centre between the needles of the other comb, they cover each other as soon as both combs are moved 1/4 inch in the opposite direction.

THE BRUSH.

To distribute the colors equally for drawn edges it is necessary to divide the color around on the size as explained in the chapter on nonpareil marble. This is almost impossible if a common brush is employed as the drops cannot be voluntarily directed by it, I have had, therefore, brushes made with bent bristles which make the distribution of the drops easier.

THE BROOM-CORN WHISK.

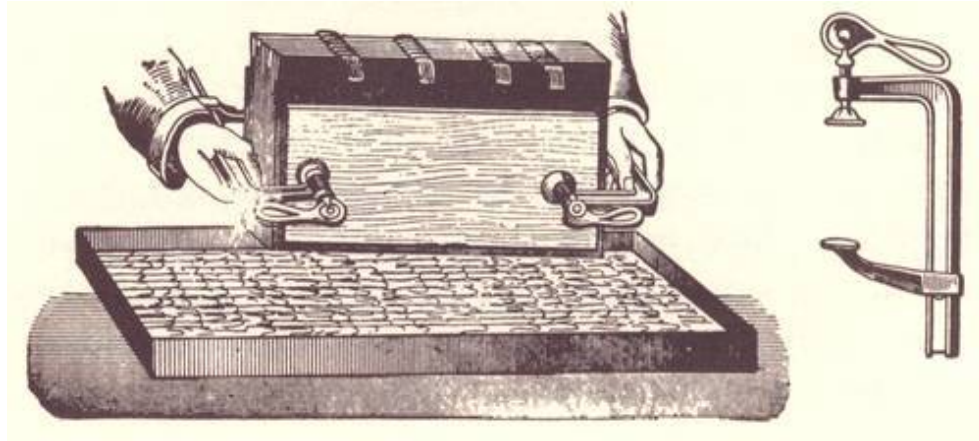
This whisk is made of soft broom-corn about 1 to 2 inches thick. The upper end is tightly bound with string or wire so that about 5 inches of straw protrude. These whisks are used in producing marble-edges.

THE BRISTLE BRUSH.

A bristle brush 3 inches long and from 2 to 2-1/2 inches thick is bound by a well waxed packing-thread about 1-1/4 inches from the top so that the bristles are open for a length of 1-3/4 inches. This brush is very well adapted for distributing the drops.

In marbling large lots of books, large broom-corn whisks are profitably used for throwing on the body color or sprinkling water. Being possessed of the above mentioned utensils which only are expensive at first, nothing is further necessary for the production of beautiful edges.

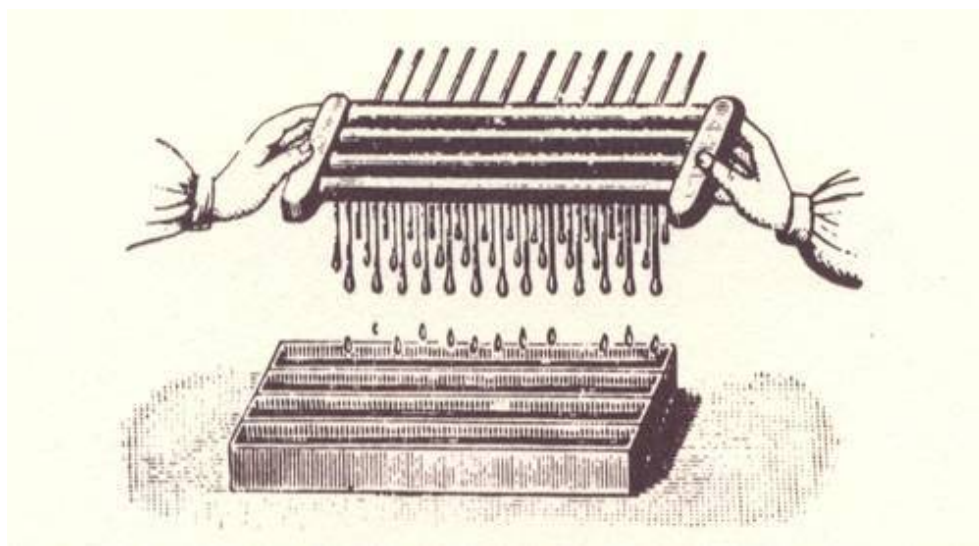
THE BOOK CLAMP.



The clamp is used to hold several books together which are to be immersed at the same time into the marbling trough. They are of iron entirely, are eccentric in their movement and make a very quick handling and an easier holding of the books possible.

APPARATUS FOR THROWING ON COLORS.

For the purpose of producing nonpareil marble an equal distribution of color is desirable.



It consists of four small color receptacles and apparatus from which protrude needles equi-distant from each other in the form of a rake.

The throwing of marble-colors at equal distances and in equally large drops on the size is made possible by it in a very certain and easy way and is even a

good help to the inexperienced marbler in the production of even and exemplary edges. For the production of comb-marble on a large scale this apparatus is especially profitable as by using it much time is saved.

Table I. TRYING OF COLORS.



Trying of Colors for Nonpareil Marble.

Black 3 to 4-1/2 inches in diameter.
Blue, yellow and red in rings within each other.

See [Page 105](#).

Table II. TRYING OF COLORS.



Size too consistent.

Result of drawing the colors with a stylus.

See [Page 104](#).

Table III. THROWING ON OF COLORS.

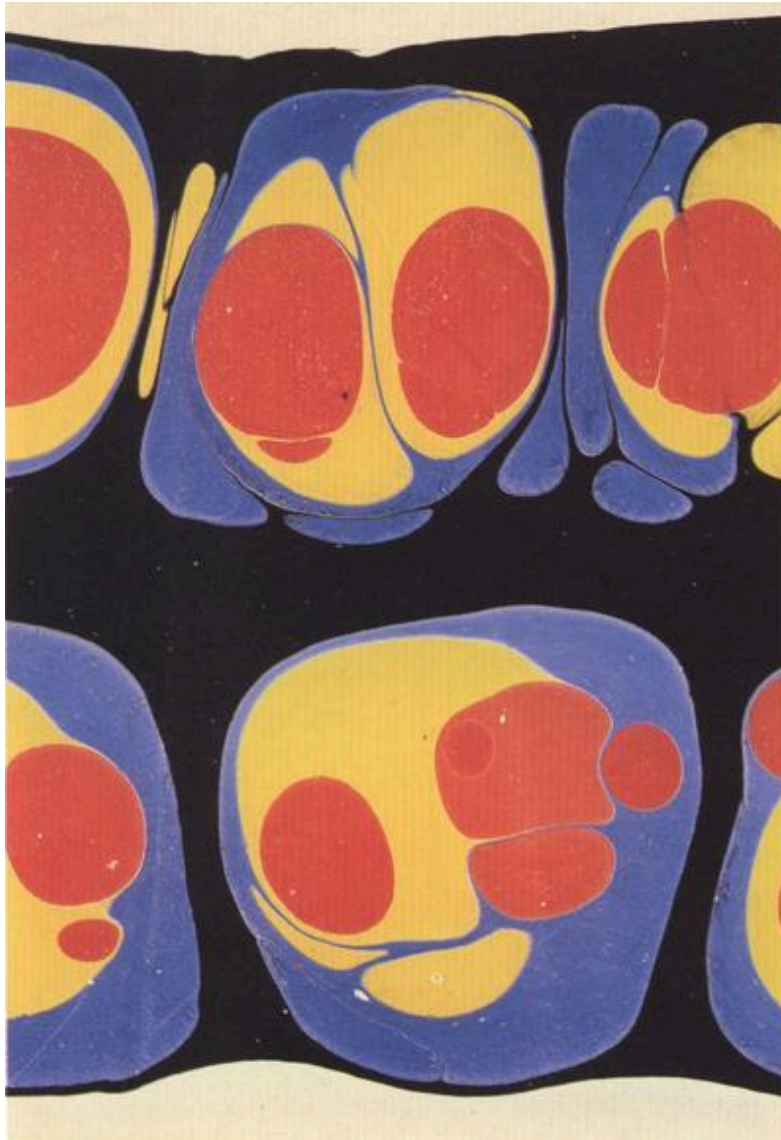


Ribbon-like Throwing on of Colors.

[First color] for Nonpareil Marble.

See [Page 105](#).

Table IV. THROWING ON OF COLORS.



Throwing on of the other colors for Nonpareil
Marble.

[Ribbon-like in two stripes.]

See [Page 106](#).

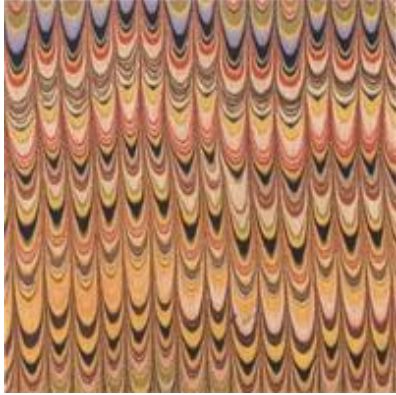
Table V. DRAWING WITH THE STYLUS.



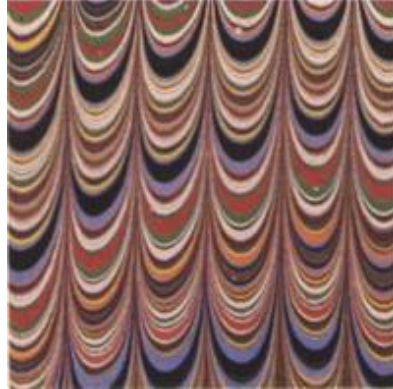
Drawing of the colors thrown on
[see preceding table] for Nonpareil or Comb Marble.

See [Page 106](#).

Table VI. NONPAREIL MARBLE.



NARROW COMB.



WIDE COMB.



NARROW AND WIDE COMB.



COMB DRAWN BACK.



DOUBLE COMB.



COMB DRAWN CROSSWISE.

Table VII. PEACOCK AND BOUQUET MARBLES.



PEACOCK MARBLE.



PEACOCK MARBLE.



PEACOCK MARBLE.



BOUQUET MARBLE.

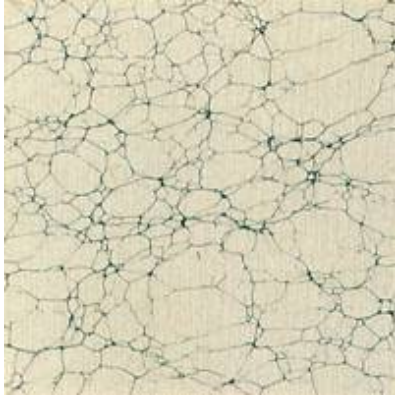


AMERICAN MARBLE.



FANCY MARBLE.

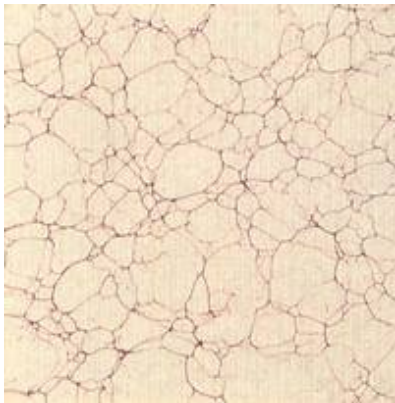
Table VIII. HAIR-VEIN MARBLE.



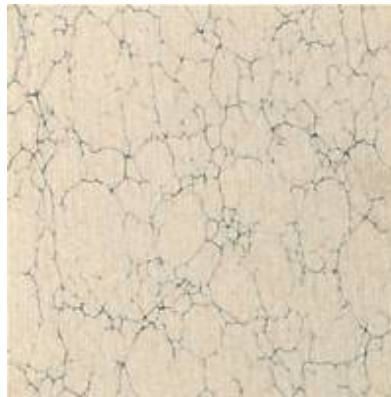
BLACK.
See [page 148](#).



BROWN.



RED.
See [page 148](#).



BLUE.
See [page 148](#).



RED AND BLACK.
See [page 149](#).

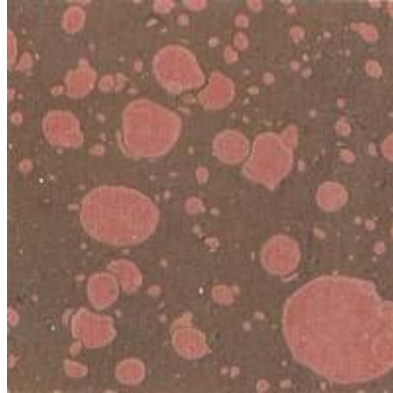


BLUISH-BROWN.
See [page 149](#).

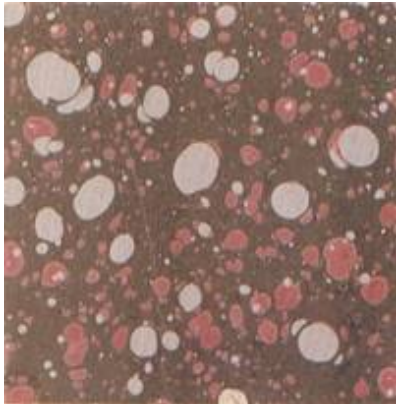
Table IX. TURKISH MARBLE.



BLACK.
Thrown on in ribbon form.
See [page 138](#).



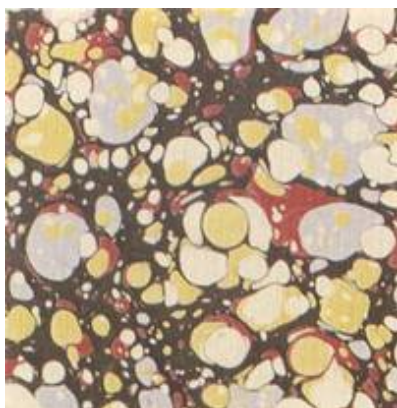
RED.
Small drops thrown on with
broom-corn whisk.
See [page 138](#).



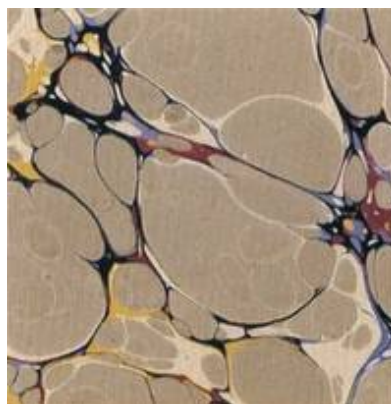
BLUE.
Small drops thrown on with
whisk.
See [page 138](#).



YELLOW.
Small drops thrown on with
whisk.
See [page 138](#).

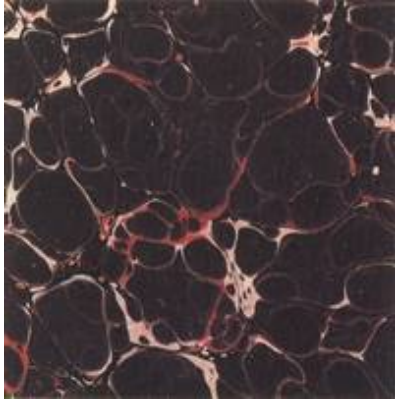


GALL WATER
to produce white veins.
See [page 138](#).

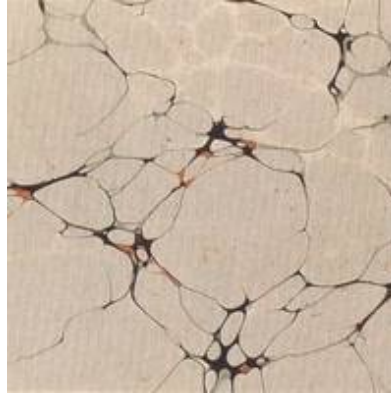


GROUND COLOR GRAY,
Sprinkling water with Black.
See pages [138](#) and [139](#).

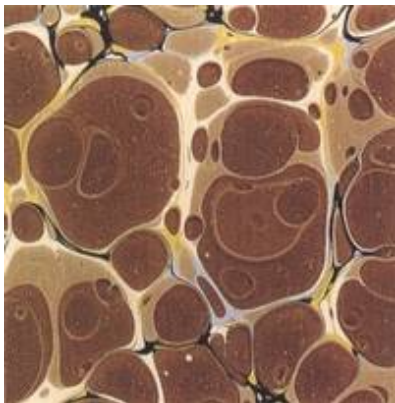
Table X. TURKISH MARBLE.



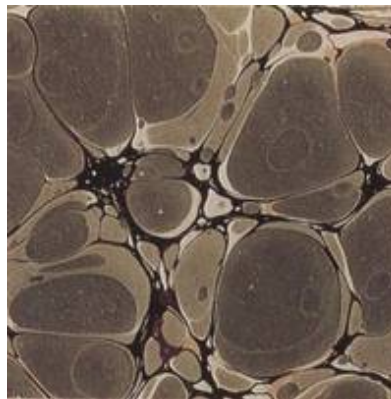
DARK RED.
See [page 139](#).



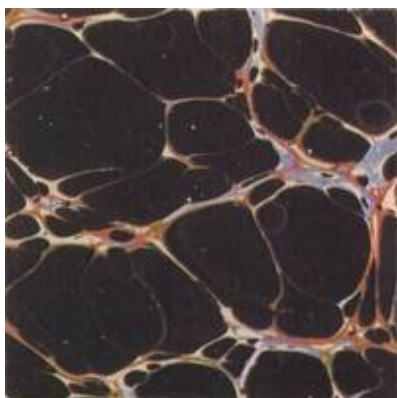
BLuish GRAY.
See [page 142](#).



BROWN.
See [page 140](#).



OLIVE.
See pages [140](#) and [141](#).



BLACK.
See [page 141](#).



CHAMOIS.



APPENDIX.

Colored AND Decorated Edges.

ACCORDING TO THE

NEWEST DISCOVERIES

BY

Eminent Specialists.

STARCHED EDGES.



STARCHED EDGES.

A great similarity to marble edges have the so called starched edges.

To produce this kind of an edge no especial practice and no important preparations are necessary, but it permits of many varieties.

For this purpose, books already cut are placed into the hand press, and in such a way, that the edge is horizontal.

Dissolve a small quantity of pure rice starch in water until it has the consistency of a paste, add the different colors, which must be soluble in water in order to color properly, such as carmine or vermilion, cinnabar, Turkey, burnt umber, Prussian-blue, ultramarine, etc.

When the color added has been carefully united by stirring with the solution of starch, throw, by the aid of a small whisk, the solution so colored in drops that are not too small on the edge placed horizontally, so that the edge is covered as uniformly as possible.

As soon as the starch thrown on is dried, sprinkle on by the aid of a sprinkling brush and sieve, a darker color over the whole surface and allow the whole edge to dry again, then brush off the superfluous starch, and the colored and spotted edge will appear with veins interspersed.

The edge will be more effective and more similar to marble, if, for example, bluish-gray colored starch and then other starch mixed with red, the former in small the latter in larger drops, are thrown on so that the edge is quite well covered. Finally, a dark-brown color is sprinkled over the whole surface.

Such edges have the effect of bluish-gray marbling imitations which are interspersed by dark veins. If you sprinkle on fluid gold bronze i. e. bronze finely ground in gum-arabic, instead of the dark-brown color, you obtain the celebrated gold veined edge in connection with the starched edge.

Unsize paper is the best for producing starch edges because it quickly absorbs the moisture of the starch and color.

To sized paper the color does not stick so readily and for that reason often runs together. To prevent this, moisten the edge before applying the starch color, with some ox-gall, and allow the first color to dry thoroughly before applying the second.

The Production of Colored Edges.



THE PRODUCTION OF COLORED EDGES.

The beautiful marbled edges have nearly driven every other kind of edges entirely out of the market.

Aside from marbled edges, colored edges are mostly used. They will keep their place on account of the simplicity of their production and their bright coloring effects.

Red-edges, especially when prettily made, give a dignified aspect to a book, so to speak, and they are therefore frequently and properly employed.

Our highly developed color industry offers to us a very great number of red colors, especially the carmine, vermilions and cinnabars, all of great value in our trade. These colors will not fade, the former carmine possesses great intensity, and cinnabar has the power to obliterate colors underlying it.

When cheap colored edges are to be produced even aniline colors can be taken into consideration. For instance, eosine is a red color, which used by itself or in combination with carmine and other red colors makes a beautiful color for edges but it will fade on exposure to the sun.

The preparation of these colors is of importance. Genuine carmine should, after the different pieces have been crushed, be ground for a short time with a little spirits of ammonia and then allowed to stand until completely dissolved. To dilute it, a thin solution of gum or water (the best is rain water) may be used. If too much spirits of ammonia has been employed the color will assume a bluish hue.

The heavy cinnabar makes a more powerful binding medium necessary, the best being a paste of starch or a solution of gum-arabic. Careful crushing and a thorough grinding of the color are essential.

Genuine carmine has a red color with a slight bluish cast whereas cinnabar approaches yellowish-red. A pleasing, pure red can be produced by using Turkish red. A bright lustrous shade is obtained by adding some red aniline color.

If edges have to be colored blue, use ultramarine blue for dark, or cobalt blue for light.

Both colors must be mixed with gum-arabic or paste by grinding, and, if necessary, be diluted with water.

To produce yellow edges, chrome yellow is mostly used. It is for sale in all different shades from the brightest greenish-yellow, to the darkest orange-yellow.

In using it, like any other mineral or earth colors, it is finely ground and mixed, for use, with paste.

Green edges are often produced by using Schweinfurt's green or silk-green. By mixing light chrome-yellow with cobalt-blue, a beautiful rich green results.

For brown and black the well known colors umber, sepia, vegetable lamp-black, drop ivory-black and others are employed.

The color for producing violet edges can be best obtained by mixing cobalt-blue with Turkish-red or carmine.

It is always the most essential part of the process to grind the color uniformly with the paste, if it is desirable to obtain an edge of the same intensity of color.

A peeling off of the color while the edges are being burnished is due to a lack of paste, or of a too great consistency or too insufficient fineness of the color.

Recently our industry has taken this article in hand and furnishes a large number of beautiful, cheap and well prepared colors for producing edges which can be bought more profitably than they can be laboriously made.

As all colored edges, to give a beautiful effect, must be burnished after coloring, it is advisable to execute the coloring while the books are in the press, place the books after their front edges have been cut, by the aid of gilding boards into the hand-press and with a soft brush apply the color thinly. If the color does not thoroughly cover on one application, the operation must be repeated after the first coat has completely dried. When a

uniform color is obtained allow the edge to dry in the press and then burnish it well with a broad burnisher.

By using a little wax you will render the burnishing more successful.

The color will sometimes enter into unsized or wood paper notwithstanding heavy pressure.

To avoid this the edges must be moistened with a thin paste to which some alum is added. It is still better to use the grounding substance manufactured by William Leo, of Stuttgart.

If beautiful edges are to be produced, then, in cutting the book it is necessary to see that the knife of the machine is free from nicks, so that the cut is clean and smooth.

At the fore-edge of the book there will always appear some small parts of the different sheets protruding slightly. If this is to be overcome, these edges are to be treated as hollow edges which means that they must be scraped hollow before coloring and must be burnished with the round-burnisher.

The use of colored edges is always left to individual taste. The following rule may be of some advantage; colored edges must have a binding of an opposite color. A red edge will appear to most advantage with a white, gray, dull-green, dark-brown or antique binding.

Bindings of a reddish-brown color would lose in their general aspect by the employment of a red edge.

If books with red edges are to be covered with red, it is then essential that the colors of both are corresponding.

GILT EDGES.



GILT EDGES.

In the production of gilt edges, there have not been any essential improvements noticed.

If the gilt-edge is to-day produced more beautifully and perfectly than formerly, it is due in part to a more extensive experience and practice, and also to the excellent quality of gold-leaf now on the market.

No edges contribute so greatly to the generally elegant appearance of a book as gilt-edges; they are lustrous and rich and are more pleasing to the eyes of most men than the plain and unseemly colored edges, though a great deal of trouble and industry is necessary to the production of these beautiful effects.

A principal condition in the production of gilt-edges is the utmost cleanliness and skill in executing the necessary manipulations.

In cutting the book, great care must be taken that the cuts are entirely smooth and clean, without any blemish. Sharp knives are therefore indispensable in this kind of work.

Gilt-edges demand an entirely smooth surface which is only produced by carefully scraping the surface with a steel scraper. The edge must be scraped perfectly smooth so as not to show the marks of the knife in cutting or those of the scraper. To this end the books, the edges of which have already been cut, are placed between gilding-boards in hand presses in such a way that the fore-edge protrudes but slightly over the cheeks of the press. After the press has been tightly drawn down and has been fastened firmly, the scraping of the edges can be begun. When this tiresome work has been finished, and an entirely smooth surface has been obtained, the surface is moistened with a paste-water and rubbed dry with clean paper-shavings. By this grinding and rubbing off, the edge becomes shiny and dry, and gains in the power of resistance, which is of great importance for the burnishing that occurs in the later stage of gilt-edging.

The smooth, hard ground of paste is not sufficient for the production of gilt-edges that are perfectly beautiful.

They must be colored lightly with bolus or chalk, (the most adapted to this work is the finely washed Armenian bole.)

The book-binders supply establishment of Wm. Leo, of Stuttgart, provides an excellent article.

This bole after it has been finely ground and scraped, must be mixed with a solution of albumen (the white of the egg) such as is used for laying on gold leaf, and must be spread lightly and evenly over the edge by a fine broad hair brush or with a fine sponge, to prohibit the scaling off of the gold.

This process must be repeated. The ground of bole gives the gilt-edge a dark shade and a high lustre and glow.

When the grounding of bole is entirely dry, and all the parts clinging to the edge have been carefully brushed off, we begin the most difficult part in the production of gilt-edges, the laying on of the gold.

As binding medium between the gold-leaf and the edge of the book, a solution of the white of the egg is used. Carefully separate the white of the egg from the yolk and dilute the same with 1 pint of water and stir until a perfect mixture of the two results. After this fluid has stood for several hours, it becomes perfectly clear, the froth of the egg floating upon the surface is removed and the solution is ready for use.

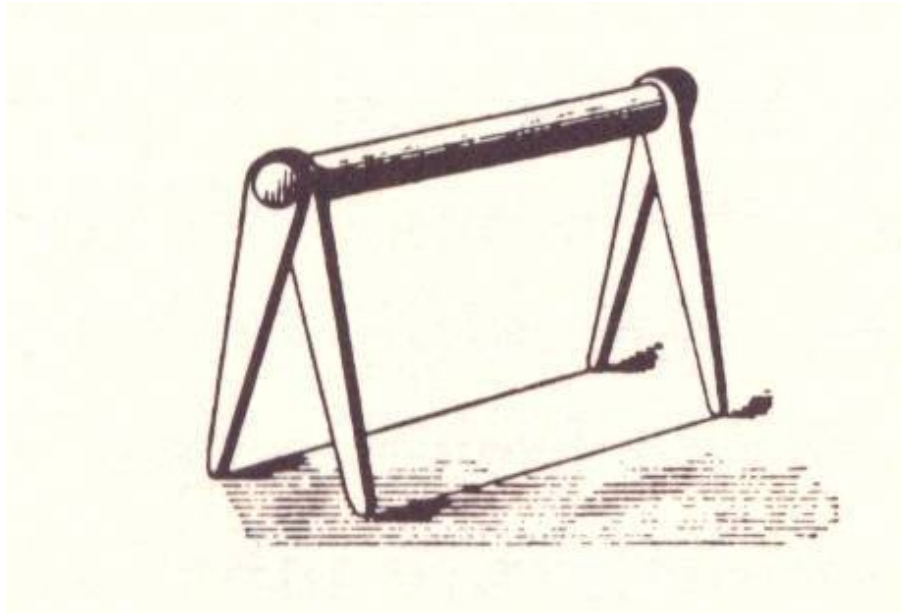
Paper containing lime makes a stronger size necessary.

To transfer the gold-leaf, which has been cut on the gold cushion to the size required, on to the edge, a rectangular light wooden frame of wood or paste board is used, across which two horse hairs or silk threads are stretched parallel to the long sides of the rectangle in such a way that they can be moved at will nearer to or farther from one another.

LAYING-ON APPARATUS.



THE LAYING-ON APPARATUS.



Much more practical than this frame is the laying-on apparatus. It can be more easily and securely handled, and the laying on strings can be more easily set.

By means of this simple instrument the sheets of gold-leaf which are laying ready on the cushion and which must be so wide that they not only cover the edge entirely but also protrude about $\frac{1}{8}$ inch beyond the gilding boards are transferred on to the sized edge. The sizing must be done with a soft camels hair brush, not too thin and it is essential that the gilding boards also become moistened.

The strings of this laying on apparatus or of the frame must be so far distant from each other that they reach from gilding board to gilding board.

The sheets of gold leaf that have been cut before must then be carefully transferred, one by one, on the sized edge.

To make the gold-leaf adhere to the strings of the frame, the strings are rubbed slightly upon the hair of the head by which means they absorb sufficient fatty substance to hold the light sheets of gold.

If after the gold is laid on it should contain any imperfections or breaks, other portions must be applied to these abrasions. The press is then placed aside until the edge is entirely dry. In treating the upper and lower edges i. e. after the book has been rounded, care must always be taken that in sizing, not too much moisture impregnates the back part of the edges as there is no pressure to prevent the size from running in.

If the gilt-edge is to have lustre and glow by burnishing, great care must be taken to dry the edge well.

An edge, which is not sufficiently dried, will exhibit breaks in the gold on burnishing, and an edge which is too dry will never have the desired lustre.

It is impossible to name the time within which gilt-edges become dry. This depends upon the temperature of the room, the quality of the paper in the book and on the quantity of albumen solution which has been put on the edge.

The safest way to find out whether the edge is dry or not is to blow upon it. If the lustre of the gold is dimmed for several seconds, the edge is still too moist, but if the cloud vanishes immediately the edge is sufficiently dry.

We can also determine by a careful easy burnishing whether the edge is dry enough to be worked further.

The burnishing should always be executed with a flat agate burnisher, and afterwards followed by a flat blood stone. The burnishing should be commenced by covering the edge with a piece of thin smooth paper that has been rubbed with wax.

First burnish the gold laid on slightly, afterwards continue with greater force.

By this the edge acquires a uniformly dull aspect but a strong union of gold and paper is thereby established.

Before beginning the real burnishing rub the edge with a soft rag upon which a small quantity of pure white wax has been rubbed. This thin film of wax renders the burnishing much easier because the burnisher is made to glide more readily and securely over the edge.

The burnishing must first be executed lightly and then with greater force.

The repairing of gilt-edges should be avoided as much as possible. Should it become, nevertheless, unavoidable, then moisten the spots to be repaired with the solution of albumen described above, lay on gold leaf, allow the spots which you wish to repair to become dry, rub off and burnish as usual.

In place of albumen solution, sulphuric ether can be used for repairing. The latter has the advantage of causing the spots so repaired to become dry immediately, but it causes a lighter spot to appear in the gilt-edge. Alcohol, lightly applied, can also be used for this purpose and with more security. The gilding of the hollow edge is produced in the same way as that of the level edge. In scraping, curved scrapers are used. To lay on the gold use the frame. Transfer the gold leaf from the gold cushion to the strings of the frame, then carefully bring these strings closer to each other, so that the gold leaf is no longer straight, but somewhat hanging down. Then transfer it to the sized edge.

To burnish give the hand press an oblique position, making the front of the press lay lower than the rear so that one half of the hollow edge has a more horizontal position, and burnish it with a flat burnisher in the same manner as the level edge. This being done, the remainder of the hollow edge is likewise treated at the same time, giving the press a different position. Only after the edge has been burnished on both sides with a flat burnisher, is the round burnisher (tooth) used to finish the hollow edge on its length.

Gilding on a colored, especially a red edge is very effective. These gilt-edges with an underlying red-edge are of English origin and are used generally for Bibles and prayer books. The fore edge of such books is colored in an oblique position of the edge, and they therefore have not a golden, but a red effect, when opened.

The production of such edges demands a certain experience, if the coloring of the fore-edge shall have a correct effect, and the leaves of the book are to be as little as possible impregnated by the color.

This color must be mixed like any other edge colors with gum-arabic, starch paste or the white of an egg.

All etching substances, such as spirits of ammonia, must be avoided as they often cause the color to soak too deeply into the paper.

In manipulating the fore edge, open the book slightly, fasten the edge in this oblique position and then lightly apply the color.

When this has thoroughly dried, open the edge towards the other side and continue with the same procedure.

The different leaves are thus not only colored at the edges proper, but the color has also been slightly imparted to the upper and lower sides of the leaves at the same time, and for this reason the edge of the book, after it is gilded, appears reddish when the book is opened out.

The upper and lower edges must be treated as all other colored edges.

In producing the gilt-edge in the way above mentioned, rubbing off with paste and grounding with bole must of course not be employed.

The production of edges with metal leaf or aluminium, is under the same rules as for gilt edges.

To produce silver edges, a sizing material of a solution of gelatine (one half of a gelatine cake to one cup of water) is used as a surrogate to the albumen. This can also be employed in the laying on of aluminium.

For base metal leaf care should be taken that a soft, thin article is used, the one of a light hue is more pliable on account of the larger quantity of tin contained in it than is in the darker copper one.

The more pliable French aluminium is preferable to the brittle German article.

If gilt-edges are to be produced upon unsized paper, the edges must be sized with a solution of gelatine.

Colored Edges with Front and End Gilding.

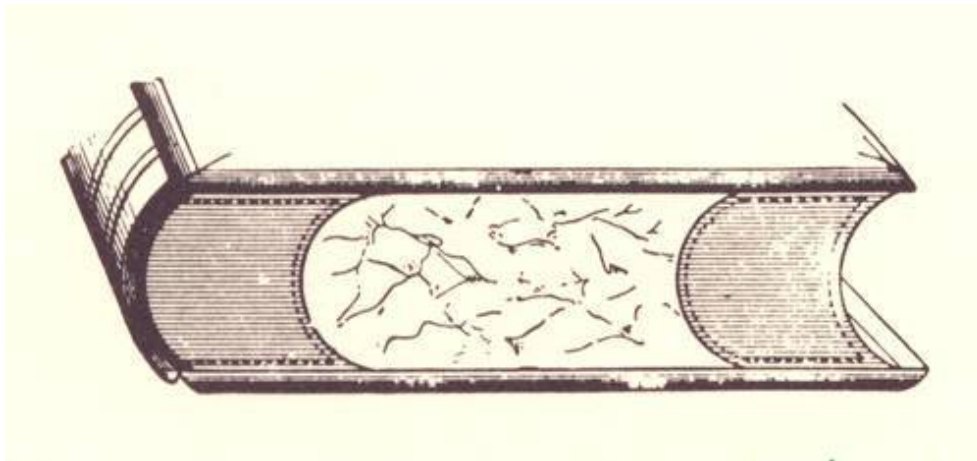


COLORED EDGES WITH FRONT AND END GILDING.

The effect of edges gilt front and back with marbling between is beautiful. This edge is especially applicable to books of large size.

The plain red-edge (carmine) and every marbled edge can be used in this connection.

Antique marbles are greatly adapted to this kind of book-edge decoration.



The production of these edges does not offer any especial technical difficulties. The book block must be treated as a hollow edge and must be scraped and then decorated with antique marbling or red color.

This having been done, insert the book into the hand-press and mark correctly how far the gilding is to be done.

To put on the gold, a solution of albumen is applied to the spots to be gilded and then a good and substantial gold leaf is laid on. As soon as the edge is dry, burnish as any other gilt-edge.

A fine chased pearl line at the meeting of the two edges, will greatly enhance the effect of this beautiful edge.

The Punched or Chased Gilt-Edge.



THE PUNCHED OR CHASED GILT-EDGE.

The procedure by which the effect of the gilt-edge is enhanced and enlivened by chasing is very old, in fact nearly as old as the gilt-edge itself. It has been driven out of the market as many another thing by over production and has finally become obsolete. In recent times it has been taken up again on account of the progress in the industrial arts and is now very highly esteemed. The design for such a decoration of gilt-edges should always be prepared by an artist and should always conform strictly to the decorations of the cover.

Technically, the production of such edges is not difficult. Some talent for drawing and a sure hand are necessary, but the artist is repaid for his labor by the interest he derives from the execution of such kind of work.

There are two ways of producing them.

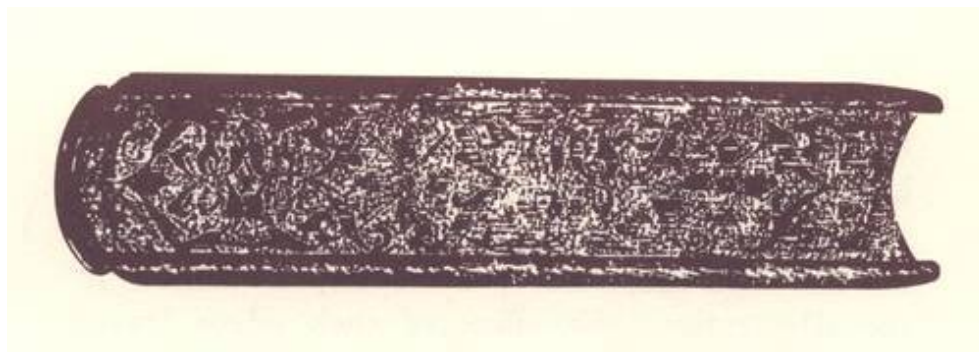
The first makes the ornament appear bright, while the ground remains dull, in the other the dull ornament appears on the shining ground surface.

The first is the more effective and the more easily produced.

The gilt-edge is here to be treated like every other plain gilt-edge, to produce dull ground, burnish with a piece of paper between the book edge and the burnisher.

CHASED EDGES.





CHASED EDGES.

Fd. Tondorf Clichè,—Property of Otto Spamer.

Transfer the design on thin tissue paper, lay the paper carefully upon the edge of the book placed in the press so that design and edge tally perfectly, and fasten the paper by pasting it on the gilding boards.

If you now by the aid of a dull awl or a pin trace the design of the tissue paper on the edge, it will appear and will be the gilder's guide. When this has been done with all necessary care, remove the paper and burnish the parts of the edge between the tracings by means of a small, pointed burnisher.

This alone relieves the ornament from the dull gilt ground.

The full effect of such edges will be imparted when the outlines of the ornaments are pressed in. To do this, fine steel stencils or stamps are best used, the same as are used for fine leather work. These stamps and the above mentioned agate burnishers can be obtained from Wm. Leo, Stuttgart.

Take the point stamp in one hand and a light hammer in the other and with an easy, equal, hammering make dot after dot of the marked pattern. If the dull ground is then profusely marked with the small pearl stamp as shown in the illustration, the effect of such edges leaves nothing to be desired.

The effect is still more surprising when the ornaments are produced in gold of a different hue or of silver or color. In the former case the ornaments are carefully penciled in with albumen solution, gold of a lighter hue, silver or aluminium is put on, and, when completely dried, is burnished.

Colored ornaments are likewise scraped out and then produced by painting.

It is a question though, whether the total effect of this last named is a refined one.

BRONZED EDGES.



THE BRONZED EDGE.

Produced by printing over colored edges. It is certainly proper to endeavor to brighten colored edges by spreading on bronzed color.

That this process has fallen more or less into disuse is probably due to the tiresome labor necessary for the making of such edges. It gives us the possibility of producing manifold changes in edges because silver or gold ornaments can be put on any color as well as on the white edge.

We have here the possibility of adapting the design of the edge to that of the cover, which under other conditions can only be done with chased edges.

It is an erroneous idea to disfigure genuine gilt-edge by putting on bronze ornaments. The base and perishable bronze can never be the correct material to brighten the effect of genuine gold.

The bronzing of edges can only be applied to colored or white edges but not to genuine gilt-edges.

For the production of bronzed edges, the following articles are necessary.

First, Engraved plate or Cliche representing the ornament with which the edge is to be decorated.

Second, Thin leaves of roller composition for the transfer of the ornament from the plate on to the edge.

Third, Roller for rolling in the gravure.

Fourth, Thick, fluid, strong amber-lacquer, as binding medium between the bronze and the color of the edge.

The following rules should be observed.

The treatment of the book block is similar to that of the hollow edge, therefore the book, after the fore edge has been cut and rounded must be scraped hollow.

As colors for the edge, the above mentioned colors, carmine, cinnabar, cobalt-blue and others are recommendable.

As binding medium a strong albumen or gum-arabic should be used. Aniline colors can be used for this purpose, but they must be dissolved in alcohol.

Such edges have a bewitching appearance if they are made in the same shade of color as the cover, but they render a certain practice in the mixture of colors necessary.

The coloring and burnishing of the edges is executed in the same manner as described in the article upon the production of colored edges. After the burnishing of the edge is executed, the time has arrived to imprint the ornament upon it.

Cut off of the thin leaf of roller composition, a strip slightly larger than the surface of the edge. Put a little of the thick, fluid amber-lacquer upon the roller, and move the roller forward and back on a smooth stone (marble slab) until the lacquer is equally spread out over the surface of the roller. When this is done, the gravure can be rolled over by aid of this roller.

The thin strip of roller composition is then carefully placed on the gravure, pressed on lightly and evenly with cotton wadding or a soft rag. By this means the ornament of the gravure is transferred to the roller composition.

To transfer this ornament on to the surface of the edge, it is necessary to lay the strip of roller composition carefully on the edge and carefully and evenly press it on.

The bronze is not to be put on to the edge on which the design is printed before the lacquer is quite dry, only then can the blurring of the ornament be avoided and the bronze be made to retain its specific metallic lustre.

After the lapse of several hours, the bronze will be dry, and then the cleaning of the edge is done with soft cotton wadding.

The strips of roller composition so used must be cleaned immediately after use with clean turpentine and, after they are dried, they can be again used.

In order to produce the necessary transfer strips use only the best English roller composition. Cut this mass into small pieces, place them in an earthen

vessel, subject them to a moderate heat until they become entirely fluid and pour the fluid mass on a piece of glass or a smooth sheet of tin. After this has cooled it should be carefully removed. To render this easier, the plate of glass or the piece of tin should be lubricated with some fine oil.

Producing Marble on Gilt Edges.



PRODUCING MARBLE ON GILT EDGES.

If gilt, silver or aluminium edges are to be marbled, they must be thoroughly dried, burnished slightly and rubbed off with a small piece of linen, silk or leather, and any blemishes or spots from which the gold or other metal has blistered must be repaired.

The dull gilt-edge is now to be moistened by means of a soft brush with a solution of powdered alum in hot water after this has formed a thin paste.

The marbling of the gilt-edge is executed in the usual way, before the layer of alum is entirely dried. After the completion of the marbling the edge must be washed with water until it is entirely clean.

After the edge has become dry, it is rubbed with a waxed rag and burnished over.

For marbling, the common marbling colors can be used.

It remains only to add that too dry edges do not take marbling colors readily.

CONTENTS.

	Page.
Preface to the first edition,	<u>3</u>
Preface to the second edition,	<u>10</u>
Historical,	<u>13</u>
The size,	<u>21</u>
Carrageen moss,	<u>31</u>
Gum tragacanth,	<u>45</u>
Salep, Flea-bane and other glutinous bodies,	<u>53</u>
The Ox-gall,—Its chemical analysis,	<u>59</u>
Effect of the gall on colors,	<u>60</u>
Preparation of ox-gall and its uses,	<u>63</u>
Substances acting similarly to gall,	<u>71</u>
Sprinkling water,	<u>75</u>
The alum water,	<u>79</u>
The preparation of colors for marbling,	<u>85</u>
The marbling of book-edges and paper,	<u>97</u>
The comb or nonpareil marble,	<u>103</u>
The peacock marble,	<u>111</u>
The bouquet marble,	<u>115</u>
The entwined comb marble,	<u>119</u>
The snail marble,	<u>123</u>
General remarks relative to drawn edges,	<u>129</u>
Marbled edges,	<u>137</u>
Hair-vein edges,	<u>147</u>
Rules in regard to the marbling table,	<u>153</u>
Precautions,	<u>153</u>
A new expanding medium,	<u>154</u>
Marbling of gilt-edges,	<u>159</u>
Marbling on a large scale,	<u>159</u>
The Utensils,—The marbling trough,	<u>163</u>

The size-skimmer,	<u>163</u>
The drawing needle or stylus,	<u>164</u>
The comb,	<u>164</u>
The peacock-comb,	<u>165</u>
The brush for comb-edges,	<u>165</u>
The broom-corn whisk,	<u>165</u>
The brush for marbled-edges,	<u>166</u>
Book-clamps,	<u>166</u>
Apparatus for throwing-on colors,	<u>167</u>
Sample plates, I-X,	<u>169-187</u>

APPENDIX.

The starched edge,	<u>193</u>
The colored edge,	<u>197</u>
The gilt-edge,	<u>205</u>
Colored edges with front and end gilding,	<u>219</u>
The chased gilt-edge,	<u>223</u>
The bronzed-edge,	<u>231</u>
The marbled gilt-edge,	<u>237</u>

FOOTNOTES:

[A] The publisher of this book, having accepted the sole agency for the United States and Canada of these colors, pledges the closest and most prompt attention to all orders, despatch in delivery and most reasonable prices.

See [price list](#) at the end of this book.



PRICE LIST OF Halfer's Imported Marbling Colors

Halfer's Superior Marbling Colors.

	5 Qt. Cans, per Qt.	One Qt. Jars.	In Pint Jars
Scarlet Red	\$1.00	\$1.25	\$.70
Carmin Lake	1.00	1.25	.70
Oriental Blue, [light]	1.00	1.25	.70
Indigo, [dark]	1.00	1.25	.70
Yellow, [lemon]	1.00	1.25	.70
Green	1.00	1.25	.70
Black	1.00	1.25	.70
Brown	1.00	1.25	.70
White	1.00	1.25	.70

SHADES PREPARED TO ORDER.

SUNDRIES.

	In 10 Lb. Lots.	Less than 10 Lbs.
Gum Hogg	\$.60	\$.65
Gum Tragacanth	.90	1.00
Irish Moss, [carrageen]	.12	.20

SUNDRIES.

	In Quarts.	In Pints.
Prepared Ox Gall	\$.50	\$.35
Sprinkling Water	.50	.35
Shellac Ammonia	.50	.35

REFINED BEESWAX, ETC., AT LOWEST PRICES.

IRON MARBLING CLAMPS.

2 inches between jaws, per dozen	\$ 5.00
4 " " " "	7.00
6 " " " "	10.00
8 " " " "	11.00

**The American
Bookbinder,
BUFFALO, N.Y.**


**The American Bookbinder,
BUFFALO, N. Y.**

Complete • Marbling • Outfits. ➤

Complete Marbling Outfits.

SIZE A.

CONSISTING OF

- 1 Marbling Trough, 14-1/2 x 23, galv. iron, japanned.
- 1 Basin for waste color, galv. iron, japanned.
- 1 Basin for sprinkling water and trying of colors, japanned.
- 1 Sprinkling Brush for hair-vein edges.
- 1 Sprinkling Sieve for hair-vein edges.
- 1/2 Dozen Broom Corn Whisks for Spot (Turkish) edges.
- 1/2 Dozen Bristol Brushes for drawn edges.
- 1/2 Dozen Color Recepticals.
- 1 Size Skimmer.
- 1 Stylus.
- 1 Narrow Comb.
- 1 Medium Comb.
- 1 Piece of Coarse Cloth for straining size.
- 6 Bottles Halfer's Marbling Colors, 1/2 pint each.
- 1 Bottle Prepared Ox Gall.
- 1 Bottle Sprinkling Water.
- 1 Bottle Shellac Ammonia.
- 1 Pound Irish Moss.
- 1/4 Pound Gum Tragacanth.
- 1/4 Pound Gum Hogg.

PUT UP IN NEAT SUBSTANTIAL BOX.

PRICE, \$25.00, NET.

SIZE B.

CONSISTING OF

- 1 Marbling Trough, 18 x 32, with out-let, galv. iron, japanned.
- 1 Basin for waste color, with out-let, galv. iron, japanned.
- 1 Basin for sprinkling water, trying of colors, galv. iron, japanned.
- 1 Sprinkling Brush for hair-vein edges.
- 1 Sprinkling Sieve for hair-vein edges.

1 Dozen Broom Corn Whisks for Spot (Turkish) edges.
 1 Dozen Bristol Brushes for drawn edges.
 1/2 Dozen Color Recepticals, holding over one pint color each.
 1 Size Skimmer.
 1 Stylus.
 1-1/8 in. Comb made of heavy press board.
 1-1/4 in. Comb, " " "
 1-1/2 in. Comb, " " "
 1 Peacock Comb, " " "
 1 Snail Apparatus made of iron and brass wire.
 1 Piece of Coarse Cloth for straining size.
 1 Large Dover Egg Beater for beating gum size.
 9 Bottles of Halfer's Marbling Colors, 1/2 pint each.
 1/2 Pint Prepared Ox Gall.
 1/2 Pint Sprinkling Water.
 1/2 Pint Shellac Ammonia.
 2 Pounds Irish Moss.
 1 Pound Gum Tragacanth.
 1 Pound Gum Hogg.

PUT UP IN NEAT SUBSTANTIAL BOX.

PRICE, \$50.00, NET.

Utensils of above Outfits may be Bought Separate at
the following Prices:

SIZE A.

Marbling Trough, 14-1/2 x 23, galv. iron, japan'd,	\$3.00
Basin for waste color, " "	.75
Basin for sprinkling water, " "	.75

Sprinkling Brush	.75
Sprinkling Sieve, brass wire	1.75
Broom Corn Whisks, each	.20
Bristol Brushes, No. 1, set in hard rubber, each	.30
Bristol Brushes, No. 2, common, each	.15
Stylus	.10
Size Skimmer	.25
1/8 in. Comb, best quality press board	3.00
1/4 in. Comb	1.30
1/2 in. Comb	1.25
Peacock Comb	2.75
Snail Apparatus, iron and brass wire	3.50
Straining Cloth, one yard square	.60
Extra large Dover egg beater, for beating gum size	3.50

SIZE B.

Marbling Trough, 18 x 32, with out-let, galv. iron, japanned	\$4.00
Basin for waste color, out-let, galv. iron, japa'd	1.00
Basin for sprinkling water " " "	1.00
Sprinkling Brush	1.50
Sprinkling Sieve, brass wire	2.50
Broom Corn Whisks, each	.25
Bristol Brushes, No. 1, set in hard rubber, each	.30
Bristol Brushes, No. 2, common, each	.15
Stylus	.10
Size Skimmer	.35
1/8 in. Comb, heavy press board	3.25
1/4 in. Comb	1.40
1/2 in. Comb	1.35
Peacock Comb	3.25
Small Apparatus, iron and brass wire	5.00
Straining Cloth, one yard square	.60
Extra large Dover egg beater, for beating gum size	3.50

P. S.—Marbling Utensils not mentioned herein, made to order at short notice and at most reasonable prices.

THE AMERICAN BOOKBINDER, Buffalo, N. Y.

Louis Dejonge & Co.,

71-73 Duane Street,

NEW YORK.

Louis Dejonge & Co.,

71-73 Duane Street, NEW YORK.

... IMPORTERS AND MANUFACTURERS OF ...

BOOKBINDERS',

POCKET ◦ BOOK ◦ AND ◦ CASE ◦ MAKERS' ◦ MATERIALS,

BOOKBINDERS',

POCKET BOOK AND CASE MAKERS' MATERIALS,

ENGLISH AND AMERICAN BOOK CLOTH,

LINEN BUCKRAM,

SURFACE-COATED, FANCY, GLAZED, PLATED,

SILVER, GOLD, AND MARBLE PAPERS,

SCRAP BOOK PICTURES,

LACES, BORDERS,

HALFER'S MARBLING COLORS.



... Try our Alderney Brand American Russia. ...



• ——— THE ——— •
American • Bookbinder
~~~~~

# THE American Bookbinder

... A Technical Journal written by Bookbinders for Bookbinders, and the ...

OFFICIAL ORGAN OF THE INTERNATIONAL BROTHERHOOD.

***PUBLISHED MONTHLY.***

**SUBSCRIPTION, \$1.25 PER YEAR.**



**DEVOTED TO THE INTERESTS OF**

## Bookbinding and Bookbinders.

The only Journal of its class published in the United States.



# TRADE LISTS.

## TRADE LISTS.

**Comprising Printers,  
Bookbinders, and Kindred  
Branches, from the very latest  
and best sources. Can Supply  
any Special Trade, or Special  
State or City....**



POST PAID.

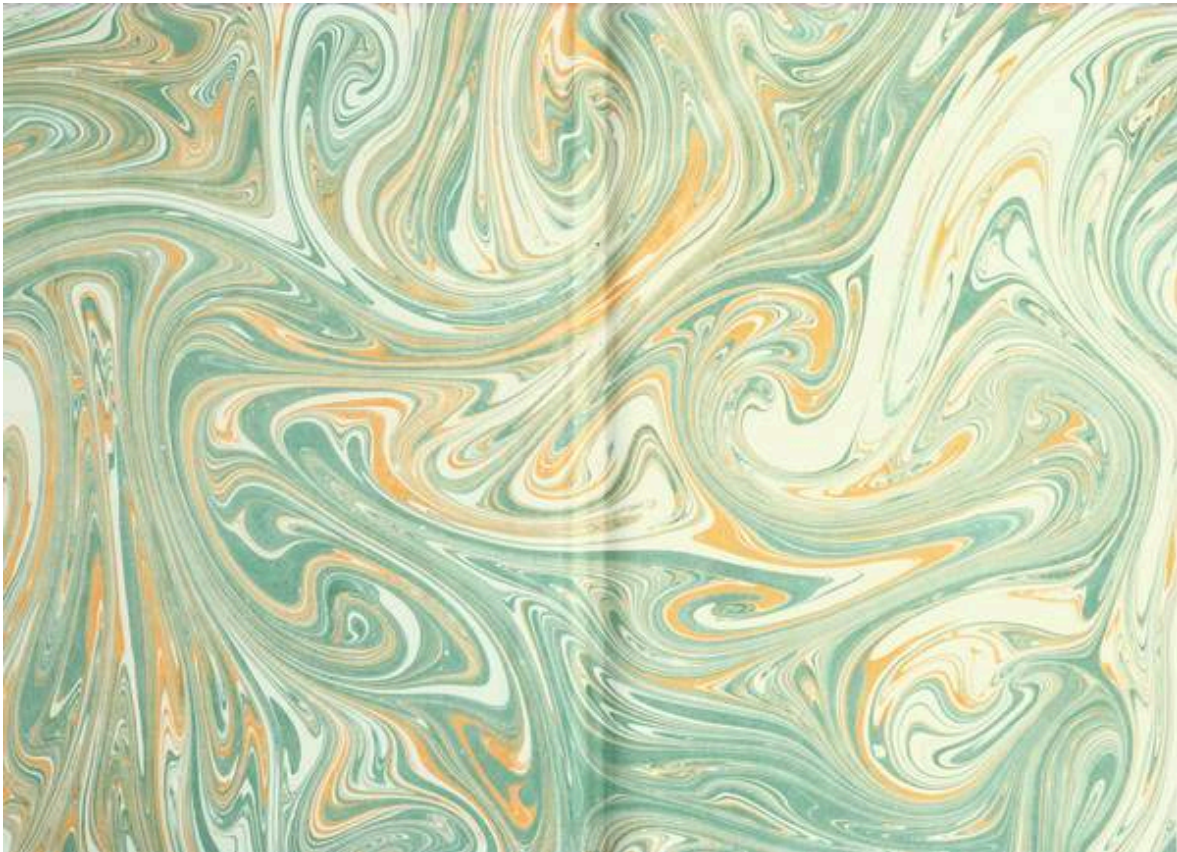
|                                                |               |
|------------------------------------------------|---------------|
| <b>Progress of the Marbling Art, Halfer,</b>   | <b>\$1.25</b> |
| <b>Nicholson's Manual, Art of Bookbinding,</b> | <b>2.25</b>   |
| <b>Zaensdorf's Art of Bookbinding,</b>         | <b>1.85</b>   |
| <b>Crane's Bookbinding,</b>                    | <b>1.10</b>   |

---

**THE AMERICAN BOOKBINDER,**

**BUFFALO, N. Y.**

---



### **Transcriber's Notes:**

- Some minor obvious typographical errors have been corrected silently.
- For the sake of consistency and clarity, fractions appearing hyphenated throughout the book have been changed from e.g. 1-4 to 1/4. Fractions were also changed from e.g. 1 1-2 to 1-1/2.
- Missing page numbers are attributed to blank pages in the original text.
- Headings in the Table of Contents do not match headings in the main body of the book.

### **Corrections made:**

- [Pg. 7](#): "are the best, is erroneous [added comma] even heavy oxides"
- [Pg. 8](#): "for color, by sufficient [replaced with "insufficient"] caution in their handling"
- [Pg. 13](#): "the extraordinary adaptibility [replaced with "adaptability"] of paper"
- [Pg. 17](#): "those indispensable [replaced with "indispensable"] to the art"
- [Pg. 32](#): "The preparation of carrageen [replaced with "carrageen"]"
- [Pg. 37](#): "saw accidently [replaced with "accidentally"] that this obstacle"
- [Pg. 38](#): "to the color to spread out. [replaced period with comma] I filled the trough"
- [Pg. 44](#): "It forms flat ribbon like [replaced with

"ribbon-like"] pieces"

- [Pg. 53](#): "yellowish white, hornlike [replaced with "horn-like" in keeping with rest of text]"
- [Pg. 60](#): "combination with tauro-cholic [replaced with "taurocholic" in keeping with rest of text] acid"
- [Pg. 60](#): "it reacts feebly [added "with"] acid"
- [Pg. 62](#): "impregnate them with alum water [added comma] allow them to dry partly"
- [Pg. 65](#): "color [added "and"] why they appear pale,"
- [Pg. 81](#): "when the excess of acetic acid is evaporated and [deleted "and" added comma] we have the desired acetate"
- [Pg. 81](#): "some [replaced with "same"] way as alum water"
- [Pg. 132](#): "bluish-green as described above [added ; here] ochre;[deleted ;] yellow, 2 parts light chrome yellow"
- [Pg. 132](#): "In producing pea-cock [replaced with "peacock" in keeping with text] marbles,"
- [Pg. 137](#): "in my many year's [replaced with "years"] experience"
- [Pg. 167](#): "a good help to an [removed "an"] the inexperienced"
- [Pg. 198](#): "use ultramarine blue, [removed comma] for dark [added comma]"
- [Pg. 213](#): "the gold laid on slightly [added comma] afterwards continue"
- [Pg. 241](#): "Irish Moss, [carageen [replaced with "carrageen"]]"
- [Pg. 242](#): "1 Sprinkling Brush for heir-vein [replaced with "hair-vein"] edges."

- [Pg. 242](#): "1 Sprinkling Sieve for heir-vein [replaced with "hair-vein"] edges."
- [Pg. 242](#): "1 Sprinkling Sieve for heir-vein [replaced with "hair-vein"] edges." (In Size B)

**Not changed:**

- [Pg. 62](#): "If 2-5 [not a fraction] ounces of rosin be boiled"

**Both versions of the following words are used in the text and have been left as they appear:**

- alkalis, alkalies
- Carrageen, carrageen
- Clichè, Cliche
- comb-edges, comb edges
- laying-on, laying on
- marsh mallow, marsh-mallows
- viz., viz

\*\*\* END OF THE PROJECT GUTENBERG EBOOK THE PROGRESS  
OF THE MARBLING ART, FROM TECHNICAL SCIENTIFIC  
PRINCIPLES \*\*\*

Updated editions will replace the previous one—the old editions will be renamed.

Creating the works from print editions not protected by U.S. copyright law means that no one owns a United States copyright in these works, so the Foundation (and you!) can copy and distribute it in the United States without permission and without paying copyright royalties. Special rules, set forth in the General Terms of Use part of this license, apply to copying and distributing Project Gutenberg™ electronic works to protect the PROJECT GUTENBERG™ concept and trademark. Project Gutenberg is a registered trademark, and may not be used if you charge for an eBook, except by following the terms of the trademark license, including paying royalties for use of the Project Gutenberg trademark. If you do not charge anything for copies of this eBook, complying with the trademark license is very easy. You may use this eBook for nearly any purpose such as creation of derivative works, reports, performances and research. Project Gutenberg eBooks may be modified and printed and given away—you may do practically ANYTHING in the United States with eBooks not protected by U.S. copyright law. Redistribution is subject to the trademark license, especially commercial redistribution.

START: FULL LICENSE

# THE FULL PROJECT GUTENBERG™ LICENSE

PLEASE READ THIS BEFORE YOU DISTRIBUTE OR USE THIS WORK

To protect the Project Gutenberg™ mission of promoting the free distribution of electronic works, by using or distributing this work (or any other work associated in any way with the phrase “Project Gutenberg”), you agree to comply with all the terms of the Full Project Gutenberg License available with this file or online at [www.gutenberg.org/license](http://www.gutenberg.org/license).

## **Section 1. General Terms of Use and Redistributing Project Gutenberg electronic works**

1.A. By reading or using any part of this Project Gutenberg electronic work, you indicate that you have read, understand, agree to and accept all the terms of this license and intellectual property (trademark/copyright) agreement. If you do not agree to abide by all the terms of this agreement, you must cease using and return or destroy all copies of Project Gutenberg electronic works in your possession. If you paid a fee for obtaining a copy of or access to a Project Gutenberg electronic work and you do not agree to be bound by the terms of this agreement, you may obtain a refund from the person or entity to whom you paid the fee as set forth in paragraph 1.E.8.

1.B. “Project Gutenberg” is a registered trademark. It may only be used on or associated in any way with an electronic work by people who agree to be bound by the terms of this agreement. There are a few things that you can do with most Project Gutenberg electronic works even without complying with the full terms of this agreement. See paragraph 1.C below. There are a lot of things you can do with Project Gutenberg electronic works if you follow the terms of this agreement and help preserve free future access to Project Gutenberg electronic works. See paragraph 1.E below.

1.C. The Project Gutenberg Literary Archive Foundation (“the Foundation” or PGLAF), owns a compilation copyright in the



collection of Project Gutenberg electronic works. Nearly all the individual works in the collection are in the public domain in the United States. If an individual work is unprotected by copyright law in the United States and you are located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work. The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg™ License included with

this eBook or online at [www.gutenberg.org](http://www.gutenberg.org). If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official

version posted on the official Project Gutenberg website ([www.gutenberg.org](http://www.gutenberg.org)), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, “Information about donations to the Project Gutenberg Literary Archive Foundation.”
- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.
- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the

electronic work is discovered and reported to you within 90 days of receipt of the work.

- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

#### 1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY

DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution

of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a) distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

## **Section 2. Information about the Mission of Project Gutenberg**

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at [www.gutenberg.org](http://www.gutenberg.org).

## **Section 3. Information about the Project Gutenberg Literary Archive Foundation**

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact links and up to date contact information can be found at the Foundation's website and official page at [www.gutenberg.org/contact](http://www.gutenberg.org/contact)

## **Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation**

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment including outdated equipment. Many small donations (\$1 to \$5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit [www.gutenberg.org/donate](http://www.gutenberg.org/donate).

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other

ways including checks, online payments and credit card donations. To donate, please visit: [www.gutenberg.org/donate](http://www.gutenberg.org/donate).

## **Section 5. General Information About Project Gutenberg electronic works**

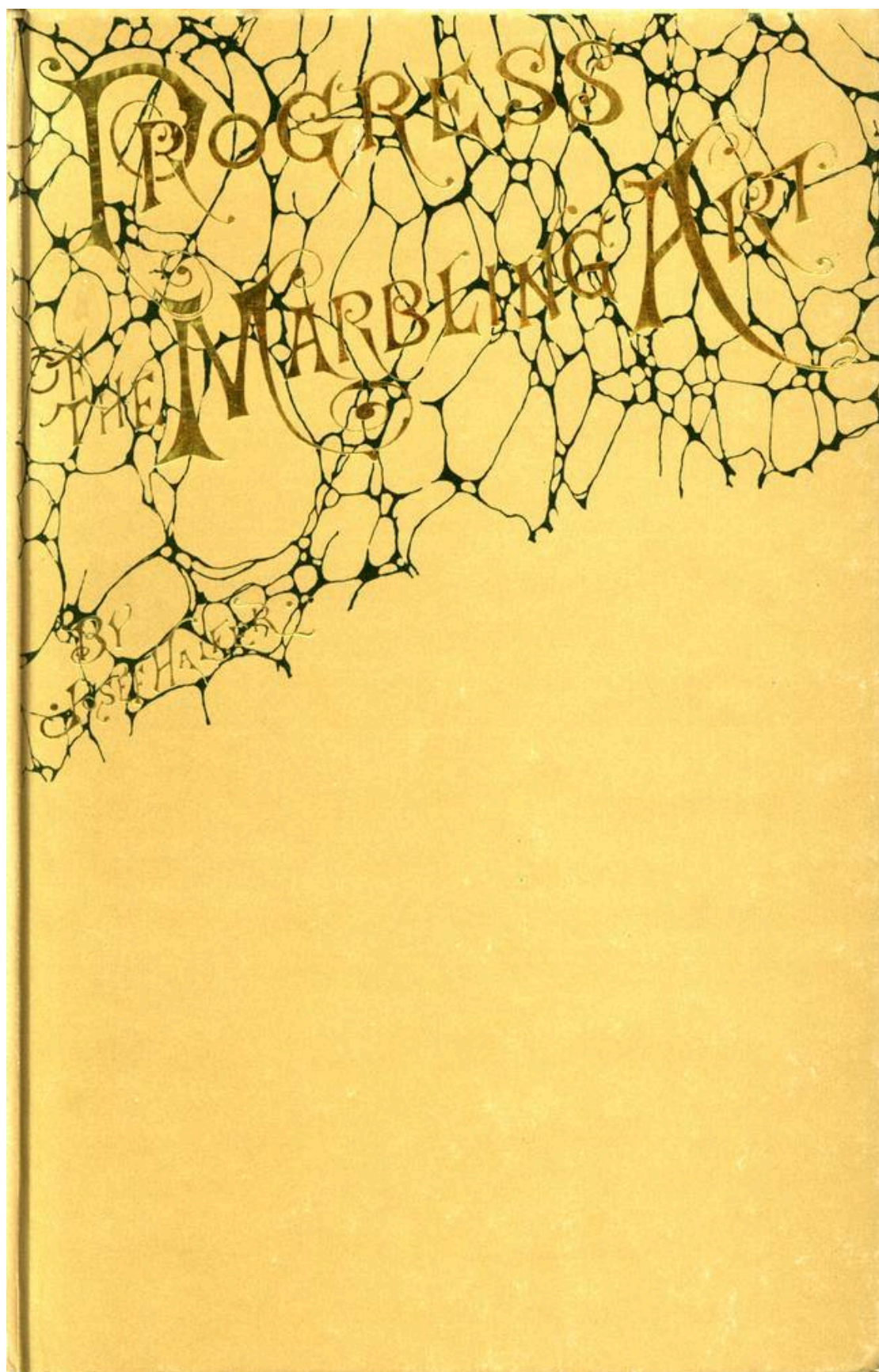
Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a copyright notice is included. Thus, we do not necessarily keep eBooks in compliance with any particular paper edition.

Most people start at our website which has the main PG search facility: [www.gutenberg.org](http://www.gutenberg.org).

This website includes information about Project Gutenberg, including how to make donations to the Project Gutenberg Literary Archive Foundation, how to help produce our new eBooks, and how to subscribe to our email newsletter to hear about new eBooks.





[back](#)





[back](#)



[back](#)